Who Bears Macroeconomic Uncertainty in the U.K.?

Theodossios Drossidis

September 2026

Abstract

Macroeconomic uncertainty recurs in every downturn, but its aggregate effects average over households with very different exposures. I trace one identified shock through the *joint* UK household distribution of income, consumption and prices, pricing each household's own basket with the published RPI group indices and carrying the distributions and their components through a functional VAR. Uncertainty compresses income and consumption, mainly because the bottom of each distribution moves up towards the middle. Benefits and spending on housing, fuel and transport compress; food spending and wages among earners do not. Prices move the other way in the first year, mainly through a fall in the price of mortgage interest, which is paid mostly by richer households. Mortgagors' consumption falls while renters' rises, with the magnitude of divergence driven primarily by higher rent payments. Uncertainty makes households more alike in what they have and, for a while, less alike in the prices they face.

Keywords: macroeconomic uncertainty, inequality, household inflation, functional VAR, mortgage channel

JEL classification: C32, D31, E31, E32

1 Introduction

Heightened macroeconomic uncertainty often coincides with economic downturns. The financial crisis, the Brexit referendum, the pandemic and the energy shock all made the future harder to forecast, and each was followed by a fall in activity. What uncertainty does to the economy as a whole is well documented: output and asset prices fall, and firms and households postpone large purchases (Bloom, 2009; Jurado et al., 2015). Who bears it is much less clear, and the answer depends on where one looks. In the United Kingdom, an uncertainty shock narrows the distributions of household income and consumption, mainly because the bottom moves up towards the middle. At the same time, it widens the distribution of the inflation rates that households face. Households become more alike in what they have and, for a while, less alike in the prices

they face. A single inequality statistic sees only the first movement, and an aggregate price index sees neither.

A growing literature asks how uncertainty redistributes. In the United States, Huber et al. (2025) find offsetting movements at the two ends of the earnings and consumption distributions, Choi and Phi (2026) show that a single inequality statistic can barely move while the gaps between rich, middle and poor households shift in opposite directions, and Fischer et al. (2021) find that income composition and labour-market fundamentals decide the sign of the response across states. For the United Kingdom, Theophilopoulou (2022) finds that inequality in income, wages and consumption falls after an uncertainty shock. Her explanation is composition: the top relies on investment income and cyclical earnings, the bottom on transfers. Splitting households by housing tenure, she also finds that mortgagors cut their consumption relative to renters. All of this evidence is about quantities. Prices enter through one aggregate index, although no household buys the aggregate basket and household inflation rates are widely dispersed (Crawford and Smith, 2002). How the distribution of household resources and the distribution of household prices respond together, for the same households, remains open.

The question matters because quantities and prices are not interchangeable. A household's inflation rate measures how the price of its basket changes; its spending records what it actually pays, housing included. For most goods the two go together. For housing they need not: what a mortgagor pays depends on the mortgage it holds, and what a renter pays on its tenancy. Studies that compare housing-tenure groups look at spending alone, often without housing costs (Cloyne and Surico, 2017; Cloyne et al., 2020). Without observing the cost of living, the housing payments and the other spending of the same households, a change in what households have cannot be told apart from a change in what their way of life costs.

This paper brings these outcomes together. The UK household survey records income, spending and the composition of the basket for the same households. I use this data to build the joint distribution of income, consumption and prices, pricing every household's own basket with the published RPI group indices from 1991 onwards. The result is a household-specific inflation rate on a payments basis, close in concept to the ONS Household Costs Indices; weighted by spending, it reproduces the published RPI. To trace how these distributions respond, I use the mixed functional VAR of Bjørnland et al. (2026), which treats a whole cross-sectional density as one variable of an otherwise standard VAR, estimated with Bayesian methods and with the uncertainty shock identified recursively. Each income source and each spending category gets its own distribution, which turns the composition argument into an estimate, and for each housing-tenure group the cost of living, the housing payments and all other spending are followed side by side.

The first result is the benchmark. An uncertainty shock compresses the distributions of gross income and consumption, mainly from the bottom. The Gini coefficient of consumption falls by 0.8 points within two years and that of income by about half a point within four. Relative to

the mean, the 5th percentile of each distribution gains five to six log points, while the 95th loses between one and one and a half. The component models show where this comes from. Benefits, the main source of income at the bottom, compress within the first year, and investment and pension income compresses from its lower end. Spending on housing and fuel and on transport compresses; food spending does not, and the distribution of wages among earners does not move at all. This confirms the composition channel of Theophilopoulou (2022) at the bottom of the distribution, where it runs through the benefit system, but not at the top. A narrower distribution of income and spending, however, is not yet a narrower distribution of living standards. That depends on what households pay.

And prices move the other way. The dispersion of household inflation widens for about three years, most of it within the first year and a half. The channel is mortgage interest. In the estimates, the 1-year gilt yield falls after the shock, and the price of mortgage interest falls with it. This cuts the inflation rate of mortgagors by up to 0.26 percentage points per quarter, while renters barely move. Mortgage holding rises steeply with income, from about 6% of households in the poorest income decile to about 69% in the richest. The cost of living turns against the poor: the poorest fifth of households faces inflation 0.14 to 0.17 percentage points above the richest fifth in the first three quarters. The raw data show the same gradient, which opens when interest rates fall. In the first year, before the compression of incomes and spending has built up, the cost of living moves in favour of richer households.

Seeing prices and payments together also changes how the tenure evidence reads. The cost of living of mortgagors falls with the price of mortgage interest, but the mortgage interest they report falls only gradually and by far less. Renters, whose cost of living barely moves, pay about 8% more rent. The total spending of mortgagors therefore falls by about 3.6% relative to that of renters after two years, the sign that Theophilopoulou (2022) finds, but mostly because renters pay more rent. Without housing costs, the measure of Cloyne and Surico (2017), the gap is about 1.3%, and valuing it at each group's own prices changes almost nothing. The cost of living, what households pay for housing and what they consume are distinct outcomes, and housing is where they part. The mortgage channel is clearly at work on prices, and it is weaker in spending.

The findings also offer a different perspective on the overall distributional effects of uncertainty. Measured in quantities, uncertainty is equalising. Measured in prices, it is not, at least in the first year. And comparisons of housing-tenure groups depend on whether housing payments count as consumption. The contribution is to measure these outcomes jointly, in one empirical framework, for the same households. Relative to Theophilopoulou (2022), the compression is located at every point of the distribution, the composition channel is estimated source by source, and the cost of living becomes an outcome rather than an aggregate control. Relative to the functional-VAR literature, the paper carries the distribution of the inflation rates that households face through the same system as their income and consumption. Through all of this, the shock changes the distances

within each distribution rather than how the three distributions line up: the rank correlations between income, consumption and household inflation barely move as such effects are not detectable here.

The rest of the paper is organised as follows. Section 2 places the paper in the literature. Section 3 builds the joint distribution from the microdata. Section 4 sets out the mixed functional VAR and its extensions. Section 5 reports the results, and Section 6 the robustness checks. Section 7 concludes.

2 Related Literature

This paper traces a single identified uncertainty shock through the joint household distribution of income, consumption and prices. It therefore speaks to two literatures that have so far developed side by side: the econometrics of macroeconomic models with functional variables, and the empirical evidence on who bears the cost of uncertainty. It also connects to work on how uncertainty affects prices. I take the three in turn.

The first literature is econometric. Macroeconomic work on heterogeneity has long summarised the cross-section with a few statistics, such as a Gini coefficient, a percentile ratio or a group mean, and placed them in an otherwise standard time-series model. This is transparent, but it decides in advance which feature of the distribution may move, and different statistics can move in different directions at the same time. The functional-data literature removes this constraint: it treats the cross-section itself as the variable to be modelled.

Inoue and Rossi (2021) took an early step by letting the *shock* be a function, a shift in the whole yield curve, rather than a single number; the cross-section as an explanatory variable still stayed outside the model. Chang et al. (2024) moved the function to the other side of the system. In their functional state-space model, the cross-sectional density of individual earnings is a state variable, estimated nonparametrically and purged of measurement error. The object of interest becomes a distributional impulse response: the change of the entire density after a shock. The framework has since been applied to many questions. Chang and Schorfheide (2024) study monetary policy and the heterogeneity of income and consumption. Chang et al. (2022) model the distribution of inflation *expectations*. Chang et al. (2025) let shocks to the income distribution feed back into the aggregates. Marcellino et al. (2024) move the functional variable to the firm side, and Ettmeier (2024) applies the same logic to tax shocks. Two recent contributions push in complementary directions. Huber et al. (2025) show that distributional responses can also be estimated by functional local projections, and Koop et al. (2026) build a pseudo-panel out of repeated cross-sections to "look within" the distribution and identify which *types* of household stand behind a given movement of the density.

The methodological anchor of this paper is Bjørnland et al. (2026). Their mixed functional VAR keeps a small block of macroeconomic aggregates and one entire distribution in the same

autoregression, truncates the infinite-dimensional part with functional principal components, and identifies structural shocks in the resulting finite system. Their application is to oil and the stock market, and methodologically, this is the estimator Section 4 uses. Bayer et al. (2026) provide a bridge from a single density to a joint one. They treat distributional data as a time series of functions with a factor structure, estimated in a Bayesian state-space model, combine micro sources of different frequency and coverage, and extend the approach to the joint distribution of consumption, income and wealth. Their aim is measurement: to reconstruct high-frequency distributions where microdata are sparse. The aim here is different. The UK survey records income, expenditure and the composition of the basket for the *same* household, so the joint distribution is observed rather than synthesised, and the spending of each household can be valued at its own prices.

A second departure concerns the variable itself. In the work above, the functional object is a quantity, such as earnings, income, consumption or firm characteristics, or a financial or expected price, such as stock returns, the yield curve or expected inflation. Here it is also the consumer price inflation that households actually face. Household-specific inflation rates, computed from each household's own basket, give a distribution of *realised* inflation, and the same estimator delivers its response alongside those of income, consumption and their components.

The second literature is empirical, and its anchor for this study is Theophilopoulou (2022), who studies the same shock in the same economy. In her UK SVAR, a macroeconomic uncertainty index in the spirit of Jurado et al. (2015) enters together with summary measures of inequality built from the same family expenditure surveys used here. Inequality in income, wages and consumption falls after an uncertainty shock, and the middle and upper part of the distribution lose most relative to the bottom: for income, the gap between the median and the 10th percentile narrows far more than the gap between the 90th percentile and the median. Her explanation is compositional: the top draws on investment income and on earnings that suffer in a downturn, while the bottom is supported by benefits. She adds housing tenure, grouping households as Cloyne and Surico (2017) do, and finds that mortgagors, who sit mostly in the middle and upper part of the income distribution, cut their consumption relative to renters.

This paper extends her analysis in four directions. First, the whole density enters the VAR instead of a Gini coefficient and two percentile gaps, so the compression can be located at every point of the distribution. It comes mainly from the bottom moving up towards the middle, which agrees with her finding that the lower half of the income distribution narrows most. Second, the composition argument is estimated rather than described. Every income source and spending category gets its own distribution and response. This shows that benefits compress with the total, that spending on housing and fuel and on transport compresses while food spending does not, and that employment income among earning households does not move at all, unlike the upper half of the individual wage distribution in her results. Third, the price side becomes an outcome: the

distribution of household inflation is modelled rather than controlled for by an aggregate index. Fourth, her tenure comparison is set against each group's cost of living and split into housing payments and all other spending, which shows that most of the relative fall in the spending of mortgagors comes from the rent payments of renters.

Huber et al. (2025) ask a closely related question for the United States with the same class of estimator, and their answer is a useful counterpoint. In their results, the share of low-earning workers first falls while the share of households reporting low consumption rises, so income inequality declines and consumption inequality increases on impact. At longer horizons the income effect reverses and the consumption effect dies out. The UK evidence below agrees with their short-run income response, since total income compresses mainly because fewer households are far below the average, but not with the reversal. Their object, however, is the earnings of employed workers, and its closest counterpart here, employment income among earning households, barely moves. Consumption compresses rather than widens, again from the bottom up.

Choi and Phi (2026) show from a different angle how much a single statistic can hide. In US data, aggregate income inequality barely reacts. Yet the gap between rich and middle-income households narrows, the gap between middle-income and poor households widens, labour income inequality rises through unemployment, non-labour income inequality falls as business and interest income dry up, and taxes and transfers compress disposable income further. Two of these movements have direct counterparts in the component responses of Section 5: the compression of investment income and of benefits. The unemployment margin has no counterpart, because the component models describe households that receive a given source. And the widening between middle-income and poor households is the opposite of what the UK data show, where the bottom moves up towards the middle. Fischer et al. (2021) reach a similar conclusion across US states: income inequality falls in most of them, and income composition and labour-market fundamentals explain the differences. The overall message of this literature is therefore fairly clear. Uncertainty compresses measured inequality, mainly because the top is exposed to capital income and asset prices while the bottom is protected by transfers. What remains open is the horizon over which this holds, the statistic in which it is reported, and the parts of the household budget that do the work.

The third strand concerns prices. The price side of the same shock has so far been studied only in the aggregate. In a panel of 82 advanced, emerging and developing economies, Beckmann and Czudaj (2026) find that higher uncertainty raises consumer price inflation, but much less, or not at all, where monetary control is strong and inflation expectations are well anchored. Whether uncertainty acts as a pure demand shock therefore depends on the monetary regime. Istrefi and Piloiu (2014) document the counterpart in expectations: in structural VARs, policy uncertainty raises short- and long-term inflation expectations even as economic activity contracts. Binder (2017) moves to the household level, though for second moments: consumers' uncertainty *about*

inflation responds to policy uncertainty, unevenly across demographic groups, and high-income and highly educated consumers are the least uncertain overall. None of these studies, however, looks at the prices that individual households actually pay.

Crawford and Smith (2002) attempt to measure such prices and construct household-specific inflation rates for the UK from the survey used below. They show that the average rate is a poor guide to the experience of most households, and that ignoring this dispersion distorts the measured growth of inequality. Their exercise is descriptive, and no shock is identified. Closest to the question asked here is Lauper and Mangiante (2025). They construct household-level inflation rates from US consumption data and show three things: a contractionary monetary policy shock lowers inflation dispersion persistently; low- and middle-income households face inflation rates that are both higher and more responsive, because of the baskets they buy; and spending on energy, water and fuel drives the result. Their paper is the closest antecedent for the inflation results below. The differences mark the contribution: a different economy, an uncertainty shock instead of a monetary policy shock, and a model of the whole distribution instead of a comparison of group means. As in their work, the link runs through the basket: expenditure shares differ along the income distribution, so an aggregate price index is the index of nobody in particular. Section 5 adds three findings. The dispersion of realised household inflation is itself a dynamic object with an impulse response. This response runs through the mortgage-interest part of the basket and the tenure split between mortgagors and renters, a channel that Cloyne and Surico (2017) and Cloyne et al. (2020) document in spending. And it can be read together with the income and consumption distributions of the very same households.

3 Data: Building the Joint Distribution

3.1 Microdata

The microdata come from the Living Costs and Food Survey, which has changed several times over the years. It started as the UK Family Expenditure Survey (FES, 1970–2001), was renamed the Expenditure and Food Survey (2001–2008) and became the Living Costs and Food Survey in 2008. Each year the survey interviews a new sample of about 5,000–7,500 households and provides survey weights. Variable codes, file layouts and the timing of the files (calendar or fiscal years) change repeatedly over the sample. I harmonise them into one consistent quarterly dataset that runs from 1977Q1 to 2020Q1 and contains 282,443 households.

Consumption is a household's total expenditure in the survey's own definition, which the ONS has kept from the FES onwards alongside the COICOP totals of the later surveys; in every year it includes housing payments, mortgage interest among them. Unless stated otherwise, income is total gross income: income before income tax and National Insurance, including benefits. Disposable income is gross income minus direct taxes. Top-coded incomes are replaced using a fitted Pareto

tail.[1] All money amounts are equivalised with the square-root scale, that is, divided by the square root of household size.

Components of income and consumption. Besides the totals, the survey records where each household's income comes from and what it spends it on. For every household the pieces add up exactly:

$$Y_{h,t} = \text{employment} + \text{self-employment} + \text{benefits} + \text{investment \& pensions} + \text{other}, \tag{1}$$

$$C_{h,t} = \text{housing \& fuel} + \text{food} + \text{alcohol \& tobacco} + \text{transport} + \text{other}, \tag{2}$$

and disposable income is $Y^{\text{disp}}_{h,t} = Y_{h,t} - \text{direct taxes}$. I check both identities household by household. "Other income" is the survey's reconciliation residual (property additions, income in kind, grants) and is negative for 12% of households. "Other consumption" collects clothing, household goods and services, personal goods and services, and leisure. Both are kept in the identities but are not modelled separately.

Household-specific inflation. The idea is the one behind the ONS Household Costs Indices, which are published for groups of UK households, and Crawford and Smith (2002) build exactly this object from the same survey. Prices are not observed for each household, but spending shares are. A household's inflation rate is therefore the official price change of each spending group, weighted by the household's own shares:

$$\pi_{h,t} = \sum_g s_{h,g,t}\, \pi_{g,t}, \qquad s_{h,g,t} = \frac{\text{spending of } h \text{ on group } g}{\text{total spending of } h}, \tag{3}$$

where $\pi_{g,t}$ is the published price index of group g. With national weights, (3) *is* the official index; with a household's own weights, it is that household's inflation rate.

I construct a household-level RPI from 1991Q1 to 2020Q1 for about 182,000 households. From 1991 onwards every survey file carries the ONS expenditure groups, the classification the ONS itself uses to weight the RPI; these groups add up to total expenditure for every household in every year. Each group is priced with its RPI group index (ONS MM23, January 1987 = 100). From 1992 onwards, housing is split further into its survey components,

$$\begin{aligned} \text{housing}_{h,t} = {} & \text{rent} + \text{mortgage interest} + \text{council tax \& rates} \\ & + \text{water charges} + \text{repairs, maintenance \& insurance}, \end{aligned} \tag{4}$$

and each component is priced with its own RPI section. This matters because rent, mortgage

[1] Appendix A describes the procedure, which follows Chang et al. (2024). Censoring affects about 1% of households overall and none before 2006, but 38% of the top decile in recent years. Section 6 repeats the three baseline models on the recorded amounts with every censored household dropped instead, and the responses are unchanged in sign, shape and timing.

interest and council tax move very differently: pricing a renter's housing with the combined housing index would wrongly expose renters to mortgage interest. In 1991, the first year of the household inflation models, the survey records an imputed rent for owner-occupiers instead of their actual payments, so housing cannot be split and is priced as one group with the combined RPI housing index.

Figure 1 shows the result. The left panel shows how far apart household inflation rates are. The right panel shows the validation: weighting households by their spending reproduces published RPI inflation. Appendix Table 1 gives the full correspondence between survey spending codes, weights, RPI indices and periods.

Crawford and Smith (2002) implement (3) in the same way, as a household Laspeyres index for the Family Expenditure Survey up to 2000. They price each household's basket with the published RPI section indices and treat housing on the same payments basis used here. Their evidence is a useful benchmark. Over their sample, only about a third of households faced an annual rate within one percentage point of the average, the average semi-interquartile range of household inflation rates was about 1.6 percentage points, and dispersion was widest when average inflation was high. Their purpose, however, is descriptive, and their series stops in 2000. The measure used here covers the overlapping decade, extends it to 2020 at quarterly frequency, and feeds the resulting densities into a VAR.

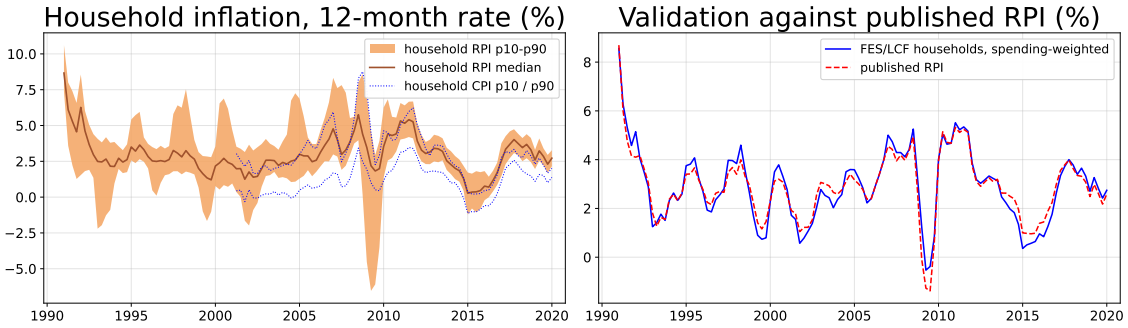


Figure 1: **Household inflation in the UK, 1991–2020.** Left: the cross-section of household-specific RPI inflation in each quarter, the shaded area between the 10th and the 90th percentile and the solid line the median, with the 10th and 90th percentiles of the household CPI measure (available from 2001) as dotted lines. Right: the spending-weighted mean of the household rates against the published all-items RPI. Rates are 12-month changes in per cent, computed from each household's own expenditure shares as in (3).

The RPI's owner-occupier depreciation, a cost based on house prices that no household actually pays, is left out. The measure is therefore on a payments basis, close to the concept of the ONS Household Costs Index but at the household level. Like the RPI and the Household Costs Indices, it treats mortgage interest as a price, whereas the CPIH values owner-occupied housing by rental equivalence and contains no mortgage interest.[2] The models use quarter-on-quarter rates in

[2]In a separate exercise I derive a CPI version at the household level (see Figure 1). That measure is only available from 2001Q2 onwards and contains no mortgage interest.

percentage points.

A joint distribution of income, consumption and prices. Every household-level file carries the same household key (survey dataset and household identifier) from the raw files onwards, and a final step joins all files on that key. For each record I check that the quarter, the survey weight and total expenditure agree across sources, and that the income and consumption identities (1)–(2) hold. The result is one record per household with income and its components, consumption and its components, and the household's own inflation rate. In total, 166,689 households are observed in all three of income, consumption and household inflation. The cross-section is therefore a genuine *joint* distribution, even where a model uses only one of its marginals.

All transformations are applied within each quarter. Money amounts, totals and components alike, enter as log deviations from the quarter's weighted mean, $\log(x_{h,t}/\bar{x}_t)$. Inflation enters in percentage points relative to the quarter's weighted mean, $\pi_{h,t} - \bar{\pi}_t$. Each variable is trimmed at the 0.5th and 99.5th weighted percentiles within the quarter. Levels, such as GDP and aggregate inflation, belong to the macro block, so the distributions describe relative positions.

3.2 Macrodata

The macro block contains a standard set of variables: real GDP growth, CPI inflation, the 1-year gilt yield and FTSE all-share growth. Uncertainty is measured by a UK macroeconomic uncertainty index constructed in the spirit of Jurado et al. (2015) at a one-quarter horizon (see Theophilopoulou, 2022). Real GDP, the consumer price index and the FTSE all-share index enter as quarter-on-quarter growth rates (one hundred times the first difference of their logarithms). The 1-year gilt yield enters in per cent and the uncertainty index in index points. The estimation sample runs from 1977Q1 to 2018Q1 (165 quarters) for income, consumption and their components; the household inflation models start in 1991Q1 (109 quarters), when the survey's expenditure groups become available. Following Bjørnland et al. (2026), all variables in the VAR are demeaned before estimation.

4 Methodology

The empirical model is the mixed functional VAR of Bjørnland et al. (2026), henceforth BCC. It treats a whole cross-sectional distribution as one variable of a VAR. This section explains the object, how its infinite-dimensional part is reduced to a few coordinates, how the dynamics are estimated and identified, and how the response of the distribution is recovered at the end.

4.1 A VAR with a distribution in it

Let $x_t \in \mathbb{R}^\ell$ collect the macroeconomic aggregates of Section 3, and let g_t denote the cross-sectional density of the household variable of interest in quarter t. Densities are non-negative and integrate to one, so the sum or the difference of two densities is not a density, and the algebra of a VAR does not apply to them directly. Following BCC, I therefore work with the demeaned density

$$f_t = g_t - \bar{g}, \qquad \bar{g} = \frac{1}{T}\sum_{t=1}^{T} g_t, \tag{5}$$

which lives in the Hilbert space $H = \{f : \int f(r)\,dr = 0,\ \int f(r)^2 dr < \infty\}$ with the inner product $\langle u, v\rangle = \int u(r)v(r)\,dr$. Stacking the aggregates and the function, $z_t = (x_t, f_t) \in \mathbb{R}^\ell \times H$, the mixed functional VAR is

$$z_t = \sum_{i=1}^{p} A_i z_{t-i} + \varepsilon_t. \tag{6}$$

This looks like a standard VAR but is not one: the A_i are bounded linear operators that map (vector, function) pairs into (vector, function) pairs, and ε_t is functional white noise. Each distribution enters its own model of the form (6), always with the same macro block.

To take (6) to the data, f_t needs a finite representation. This takes two steps, measurement and projection, followed by a choice of basis. The first step measures the density. In each quarter, the density is estimated from the n_t surveyed households with a survey-weighted Gaussian kernel density estimator and Silverman's bandwidth. It is evaluated on one grid $r_1 < \cdots < r_M$ that is common to all quarters. The function is thus represented by the vector $(g_t(r_1), \ldots, g_t(r_M))'$, and integrals become Riemann sums.[3] As BCC note, the number of grid points is a matter of numerical precision, not a parameter of the model: results do not change as long as it is not too small.

The second step projects the density onto a few basis functions. Let $(v_k)_{k\geq 1}$ be an orthonormal basis of H, and let P_m denote the projection on its first m elements,

$$P_m f_t = \sum_{k=1}^{m} c_{kt}\, v_k, \qquad c_{kt} = \langle f_t, v_k\rangle. \tag{7}$$

Replacing f_{t-i} in (6) by $P_m f_{t-i}$ leaves an approximation error $A_i(I - P_m)z_{t-i}$, which becomes negligible when m grows with T at a suitable rate (BCC, Section 2.3.1). The projection therefore replaces the whole function by m coordinate series $(c_{1t}, \ldots, c_{mt})$. The entire distribution enters the analysis through these m numbers per quarter, and the operators A_i of (6) become ordinary matrices (see Section 4.2 for the resulting finite system).

As the basis, I use the functional principal components (FPCs) of (f_t): the eigenfunctions of the sample covariance operator $\Gamma = T^{-1}\sum_t f_t \otimes f_t$ with the largest eigenvalues. On the grid, this is simply principal component analysis of the $T \times M$ matrix of demeaned densities. BCC show

[3] Appendix C.1 sets out the estimator and the construction of the grid.

that this data-driven basis is optimal in a least-squares sense, and that for a given m it captures the movements of the densities over time far better than moment, quantile or histogram bases. Two properties matter for what follows. First, each FPC is a linear combination of the f_t and therefore integrates to zero: the model can only move probability mass around, never create it. Second, the FPCs $v_k(r)$ are fixed shapes over the support of the variable, while the loadings c_{kt} change over time and carry the dynamics. I set $m = 3$. Figure 11 in Appendix C.2 shows that three components explain 87%, 91% and 91% of the variation of the consumption, income and household inflation densities, and that a fourth adds only three to five percentage points. Because the survey is not seasonally adjusted, the loadings are taken net of quarter dummies before estimation.

4.2 Dynamics and identification

Collecting the aggregates and the coordinates (the factors) of the projected density in

$$Z_t = (x_t', c_{1t}, \ldots, c_{mt})' \in \mathbb{R}^n, \qquad n = \ell + m, \tag{8}$$

the projected version of (6) is a conventional finite-dimensional VAR with distributional factors,

$$Z_t = \sum_{i=1}^{p} \tilde{A}_i Z_{t-i} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma), \tag{9}$$

with $p = 4$ lags and $n = 8$ variables: the five macro series of Section 3 and $m = 3$ loadings, all demeaned. From here on everything is standard VAR machinery; the distribution enters only through the three loadings.[4]

Estimation. I estimate a Bayesian version of the BCC model above. Here, I use the asymmetric conjugate prior of Chan (2022). It is a Minnesota-type prior that shrinks the lags of other variables much more strongly than a variable's own lags, and it keeps the posterior available in closed form, equation by equation. The posterior can therefore be sampled directly: I take 5,000 independent draws and replace explosive ones, so no burn-in is needed. All bands reported below are posterior credible bands computed draw by draw. Appendix C.3 sets out the prior in full.

Identification. The reduced-form innovations are mapped into structural shocks ε_t by

$$u_t = B\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, I_n), \qquad \Sigma = BB', \tag{10}$$

where B is the lower-triangular Cholesky factor of Σ. The impact matrix is therefore triangular in the order in which the variables enter. I place the macroeconomic block first, the loadings of the distribution next and the uncertainty index last: real GDP growth, CPI inflation, the 1-year

[4] The model is close to the familiar static FAVAR, with the factors obtained by principal components on the grid on which the smoothed densities are evaluated.

gilt yield, FTSE all-share growth, the loadings c_{1t}, c_{2t} and c_{3t}, and the uncertainty index (see also Theophilopoulou, 2022). The identified uncertainty shock is thus the part of the innovation to uncertainty that is unrelated to the contemporaneous macroeconomy and to the contemporaneous distribution. This is the conservative choice for the question asked here: the distribution cannot respond on impact, and every contemporaneous co-movement is attributed to the other variables. Responses are normalised to an increase of 0.15 index points on impact, one standard deviation of the shock.

Reconstructing distributional responses. The structural responses of (9)–(10) to the uncertainty shock include the responses $\psi_{k,h}$ of the m loadings at horizon h. Mapping them back through the basis gives the functional impulse response function (FIRF)

$$\Delta f_h(r) = \sum_{k=1}^{m} \psi_{k,h}\, v_k(r). \tag{11}$$

This is a *function* at every horizon: positive where the shock adds probability mass and negative where it removes it. Because every FPC integrates to zero, so does Δf_h: the shock moves households around the distribution rather than adding or removing them.

To express the response in familiar statistics, the FIRF is added to a baseline density, here the last quarter of the sample:

$$g_h^{\text{shock}} = g_T + \Delta f_h. \tag{12}$$

By construction, g_h^{shock} integrates to one. It is not guaranteed to be non-negative; as in BCC, the rare negative values in the tails are set to zero and the density is renormalised. Any statistic $S(\cdot)$ of the distribution can then be read off as $S(g_h^{\text{shock}}) - S(g_T)$, draw by draw, so that the reported bands are posterior bands of the statistic itself. I report the cross-sectional standard deviation, the one statistic defined for all three variables, together with Gini coefficients for income and consumption, percentiles and quantile spreads. The Gini coefficient is not defined for household inflation, which is a percentage-point deviation and takes negative values, so quantile spreads are used instead.

4.3 Beyond the marginals

Two further exercises help to detect the economic narrative behind the findings shown with the model in Section 4.2, both with the same identification approach.

Budget shares. Household inflation is, by (3), a share-weighted sum of group price changes, so its response splits exactly into group contributions: $\Delta\pi_{h,h'} = \sum_g s_{h,g}\, \Delta\pi_{g,h'}$. I therefore estimate a second VAR in which the seventeen RPI price series replace the loadings, and map each posterior

draw of the price responses through the spending shares of individual households.[5] This decomposition imposes the budget identity through a standard, aggregate VAR, whereas the functional model imposes no such structure on how households are exposed to prices. Comparing the two shows how much of the distributional response this price exposure can explain, and how size, shape and timing differ.

Housing tenure and income groups. Section 5.3 compares groups of households. By housing tenure, mortgagors pay mortgage interest, renters pay rent and no mortgage interest, and outright owners are the rest; by income, the groups are the five quintiles of equivalised gross income. All groups are formed within each quarter, from 1992Q1, when the survey first splits housing payments into their components. Consumption is measured in two ways, both per capita, as in the literature. *Total* consumption includes housing payments. *Non-housing* consumption leaves out rent, mortgage interest, council tax and rates, water charges and repairs, maintenance and insurance; it is the "total non-housing expenditure" of Cloyne and Surico (2017). Following them, each measure is trimmed at its 1st and 99th percentiles within each quarter and group. For a group $\mathcal{J}$ in quarter t, let $\bar{C}_{\mathcal{J},t}$ denote the survey-weighted mean of one of the two measures over its households and $\bar{\pi}_{\mathcal{J},t}$ the survey-weighted mean of their quarter-on-quarter inflation rates on the same basket, in percentage points: the full basket of (3) for total consumption and its non-housing part for non-housing consumption. $\bar{C}_t$ and $\bar{\pi}_t$ are the same means over all households. Each group has two series, its consumption and its inflation relative to all households,

$$C^{\text{rel}}_{\mathcal{J},t} = 100 \log \frac{\bar{C}_{\mathcal{J},t}}{\bar{C}_t}, \qquad \pi^{\text{rel}}_{\mathcal{J},t} = \bar{\pi}_{\mathcal{J},t} - \bar{\pi}_t, \tag{13}$$

both net of quarter dummies. Deflating the spending of every household by the same price index, such as the published RPI or CPI, divides $\bar{C}_{\mathcal{J},t}$ and $\bar{C}_t$ by the same number and leaves $C^{\text{rel}}_{\mathcal{J},t}$ unchanged. $C^{\text{rel}}_{\mathcal{J},t}$ is therefore relative consumption at a common deflator, whatever that deflator is. Deflating each group by its own prices instead subtracts the group's price level relative to all households, which is its cumulated relative inflation:

$$P^{\text{rel}}_{\mathcal{J},t} = \sum_{s \le t} \pi^{\text{rel}}_{\mathcal{J},s}, \qquad C^{\text{rel}}_{\mathcal{J},t} - P^{\text{rel}}_{\mathcal{J},t} \text{ is relative consumption at own prices.} \tag{14}$$

Each group and measure has its own VAR in real GDP growth, CPI inflation, the 1-year gilt yield, FTSE all-share growth, $C^{\text{rel}}_{\mathcal{J},t}$, $\pi^{\text{rel}}_{\mathcal{J},t}$ and the uncertainty index, in this Cholesky order and with the specification of the main models. The total spending of mortgagors and of renters is also split into the housing payment that defines the group, mortgage interest or rent, and all other spending,

[5] The income-quintile and tenure means use the spending shares of the households observed in the last four quarters of the sample, 2017Q2–2018Q1, and the p90 − p10 spread those of the last quarter. The functional responses are read off the distribution of the last quarter, (12), so the decomposition applies the shock to the baskets households hold at that point; four quarters rather than one average out the seasonality of spending and quadruple the number of households.

each relative to $\bar{C}_t$. Each part enters a VAR of the same form together with its own inflation rate relative to $\bar{\pi}_t$: the price change of its RPI section for the housing payment, and the inflation of the rest of the basket for all other spending. Two further VARs use the gaps between mortgagors and renters and between the poorest and the richest quintile directly. Because the VAR is linear, the response of relative consumption at own prices is the response of $C^{\text{rel}}_{\mathcal{J},t}$ minus the cumulated response of $\pi^{\text{rel}}_{\mathcal{J},t}$, computed draw by draw.

5 Empirical Analysis

All responses below are to the same identified shock: an increase of 0.15 index points in macroeconomic uncertainty on impact. Figures show 68% and 90% posterior credible bands; numbers quoted in the text refer to the 68% bands. The functional models use the settings of Section 4: three functional principal components, four lags, the macro block of Section 3 with uncertainty ordered last, and 5,000 posterior draws. The distributions of income, consumption and their components are estimated from 1977Q1 to 2018Q1 (165 quarters). The one exception is transport: the survey reports no household total for it between 1988 and 1990, so its model starts in 1991Q1. Household inflation starts in 1991Q1 (109 quarters), when the survey's expenditure groups become available, and the comparison of housing-tenure groups in 1992Q1 (105 quarters).

Because every household variable is measured relative to its own quarterly mean, the responses describe *relative* positions. A statement such as "mass leaves the lower tail" means that fewer households are far below the average household, not that anybody is better off in levels. The survey is also a repeated cross-section, so all such statements describe the distribution in each quarter, its quantiles and its within-quarter ranks, not the same households followed over time. The effects on the level of activity are in the macro block. The two can be combined, and Appendix D.1 does so: adding each distribution's own quarterly mean to the VAR leaves the relative responses essentially unchanged and delivers the response of every percentile in levels (Figure 14).

5.1 The macroeconomic response

Figure 2 shows the aggregate responses in its first five columns. They reproduce the familiar picture of an uncertainty-driven slowdown. Real GDP growth falls by half a percentage point at a one-year horizon, the same peak response as in Theophilopoulou (2022). FTSE all-share growth falls by about four percentage points after two quarters, and the 1-year gilt yield declines by 0.7 to 0.9 percentage points over two years. Aggregate CPI inflation rises by about a third of a percentage point at one year in the full sample. The uncertainty index itself keeps rising for two to three quarters and returns to zero after about three years. In the post-1991 sample used for the inflation models, the pattern is comparable: the output response is similar, CPI inflation shows a somewhat muted response, and the gilt yield falls by more, by about 1.3 percentage points after two and a

half years.

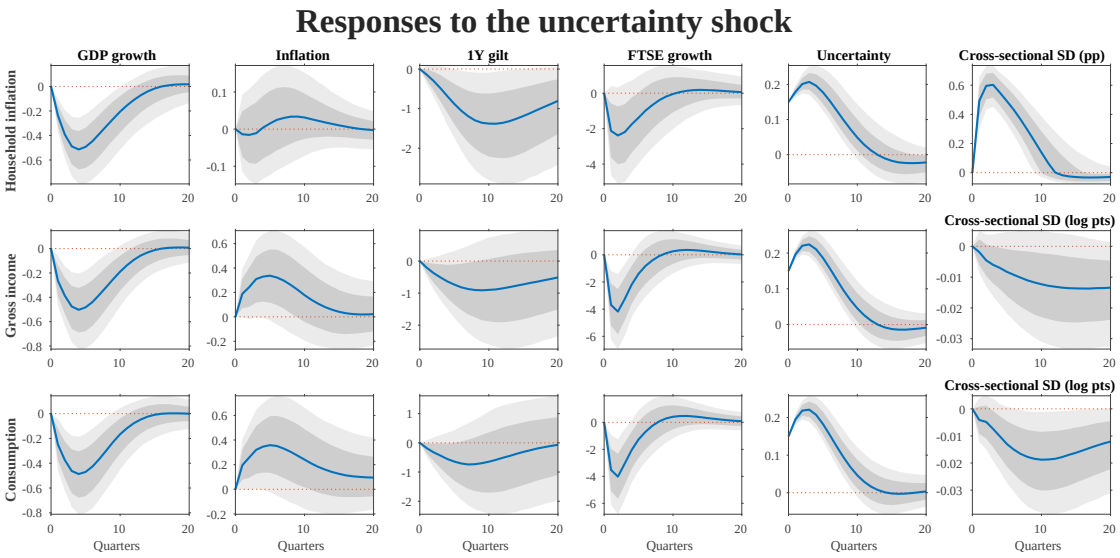


Figure 2: **Responses to the uncertainty shock.** Each row is a separate mixed functional VAR: household inflation, gross income and consumption. The first five columns show the macroeconomic block, which is the same in every model (GDP growth is the growth of real GDP), and the last column the response of the cross-sectional standard deviation of that distribution (percentage points for inflation, log points for income and consumption). Posterior medians with 68% (dark) and 90% (light) bands, per shock of +0.15 index points on impact.

5.2 The distributional response

The last column of Figure 2 summarises each distribution by its standard deviation. Figure 3 shows where in each distribution the mass actually moves, and Figure 15 in the appendix reads the same responses off five points of each distribution, from the 5th to the 95th percentile. The discussion below explains what lies behind these responses.

Household inflation. Figure 2 shows that the dispersion of household inflation rises sharply in the first quarter after the shock, peaks within the first year and stays above zero for about three years. Figure 3 shows where this comes from: mass leaves the middle of the distribution and piles up in both tails, most visibly in the first year. The percentiles in Figure 15 put numbers on it. The 95th percentile rises by about one percentage point and the 5th falls by about three quarters of a point, both peaking two to three quarters after the shock, so the gap between them widens by about 1.7 percentage points. The 25th and 75th percentiles, by contrast, move by less than a tenth of a point each. The widening is therefore a tail phenomenon.

Figure 4 shows the results of budget-share decomposition of Section 4.3 by mapping the IRFs of the seventeen RPI price series in Figure 16 through the household spending shares. The left panel compares the implied change in the p90 − p10 spread with the same statistic from the functional model. The centre and right panels average the implied household responses within

income quintiles and within tenure groups, so each line is a group mean of a share-weighted sum of aggregate price responses.[6]

The mechanism is visible in the price series themselves. The RPI mortgage-interest index falls by between 2.3 and 3.6 per cent per quarter over the first year, by far the largest of the seventeen responses, alongside the decline in the gilt yield in Figure 2.[7] Because this series enters only the baskets of households that pay mortgage interest, the right panel of Figure 4 splits the population sharply. The inflation rate of mortgagors falls by 0.24 to 0.26 percentage points per quarter over the first three quarters and returns to zero by the end of the second year.[8] Renters barely move, and outright owners see an increase of up to 0.2 points in the second year. At its peak, the gap between mortgagors and renters is about -0.28 percentage points.

Figure 17 in the appendix shows why the three groups part ways. It compares the average basket of each tenure group over 1992–2018. Mortgage interest takes 12.6% of the spending of mortgagors and nothing from the other two groups. Renters spend 21.6% of their budget on rent instead, and outright owners pay neither rent nor mortgage interest. What the groups buy besides housing then decides how their prices move. After the shock, the prices of food and catering, leisure services and household goods rise, and those of motoring follow after a short dip, while the RPI rent index falls slightly (Figure 16). Outright owners spend more than either of the other groups on food, leisure services and household goods, and they have no mortgage interest or rent whose price could offset the rise, so their inflation rate goes up. Renters spend less than either group on leisure services, household goods and motoring, and more than a fifth of their budget on rent, so the rises and the fall roughly cancel and their inflation rate barely moves.

Figure 5 shows that tenure, in turn, is strongly linked to income. The left panel splits each income decile into renters, mortgagors and outright owners, averaged over 1992–2020. Mortgagors are about 6% of the bottom decile and about 69% of the top, while renters and outright owners make up most of the lower half of the distribution. The right panel shows that this gradient holds throughout the sample: the share of mortgagors in the top decile drifts down from about three quarters to about three fifths, but it stays far above the share in the middle and the bottom decile. The same mechanism therefore produces the income gradient in the centre panel of Figure 4: the poorest quintile faces inflation 0.14 to 0.17 percentage points above the richest in the first three quarters. After about two years the ranking reverses slightly, and the poorest quintile ends up below the rest.

[6] See also Lenza and Slacalek (2024), who map aggregate responses into household-level distributions in a comparable way for income and wealth after a monetary policy shock.

[7] How closely mortgage interest follows market interest rates depends on the share of variable-rate mortgages and on how far rates can fall. Both changed over the sample, with rates close to their floor after 2009 and fixed-rate mortgages more common, so the estimates describe the average strength of the channel over 1991–2018 rather than its strength today.

[8] This is the fall in their cost of living as the RPI and the Household Costs Indices measure it, for an unchanged mortgage. It differs substantially from the change in the mortgage interest that mortgagors report in the survey: in a VAR of the same form as in Section 5.3, reported mortgage interest falls by about 4% relative to average spending after two years, with a band that includes zero, against about 23% for the cumulated RPI mortgage-interest index. The two capture different objects, the price of servicing a given mortgage and the payments on the mortgages households actually hold, and Section 5.3 treats them as separate outcomes.

Figure 6 shows that the gradient is not an artefact of the model. The upper panel plots the raw data quarter by quarter: the 12-month household inflation rate of each income decile relative to the mean of that quarter, with red cells marking deciles that face above-average inflation and blue cells deciles that face below-average inflation. The colours do not stay put. In some periods the bottom deciles are red and the top deciles blue, in others the reverse. The lower panel, on the same time axis, condenses the heat map into one line, the gap between the poorest and the richest decile, and adds the change in the 3-month Treasury bill yield and in the 1-year gilt yield over the previous twelve months. The gap opens when rates come down and closes or reverses when they go up. It was widest in 2009, at about six percentage points, when the Treasury bill yield fell by more than four points within a year; in the upper panel, this is when the top deciles turn deep blue and the bottom deciles deep red. This pattern is consistent with the cash-flow channel of interest rates documented by Cloyne et al. (2020), seen here in the *cost of living* rather than in spending. It sits next to the finding of Theophilopoulou (2022) that mortgagors cut their consumption relative to renters after an uncertainty shock, and it explains why an index that includes mortgage interest, such as the ONS Household Costs Index, diverges systematically across household groups. Section 5.3 sets the two side by side.

The decomposition also speaks to magnitude through the left panel of Figure 4, which compares the two approaches on the spread between the 90th and 10th percentiles. In the functional model, the spread widens by about 0.9 points two to three quarters after the shock and is back at zero after about two and a half years. With the budget identity imposed, it widens by at most about 0.28 points, reached in the third to fifth quarter, and fades more slowly. The two approaches therefore agree on the sign and the channel, but not on size or timing: the functional model puts much more of the response in the far tails and concentrates it in the first year.

The decomposition constrains the household responses through the budget identity: households can only differ through their exposure to aggregate price movements. The functional model imposes no such structure. Comparing the two thus shows how much of the distributional response differences in price exposure can explain. One possible reason for the larger but shorter-lived functional response is that it also picks up short-run movements in the distribution that the fixed mapping from expenditure shares to aggregate prices cannot capture. I therefore read the shape and the channel from the functional model and treat its magnitudes as approximate, a caution that Section 6 reinforces.

Household income. Figure 2 shows that the dispersion of gross income falls steadily and slowly. Figure 3 shows that mass leaves the lower tail, around the 5th percentile, and to a much lesser extent the upper tail, and gathers in the lower half and the middle of the distribution; the picture takes two to four years to form. Figure 15 shows the same picture through the percentile IRFs. Relative to the mean, the 5th percentile gains about six log points by the third year and the 25th nearly two, while the 75th and 95th lose at most about one. The Gini coefficient in Figure 18

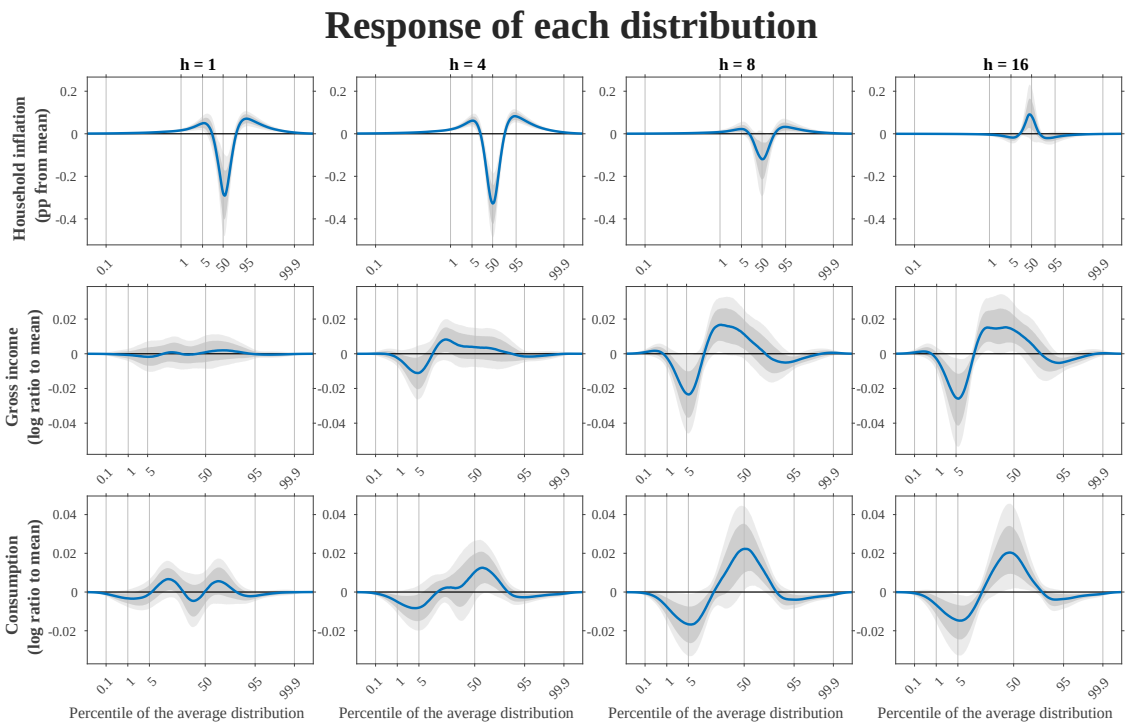


Figure 3: **The response of the distribution.** Functional impulse responses (11) at horizons $h = 1, 4, 8, 16$, by row: household inflation, gross income and consumption. A positive value means that the shock puts more mass at that point of the distribution. The horizontal axis is marked in percentiles of the time-average density, the object the demeaned density of (5) deviates from, and each panel is cut just outside the 0.1st and 99.9th percentiles. Posterior medians with 68% and 90% bands.

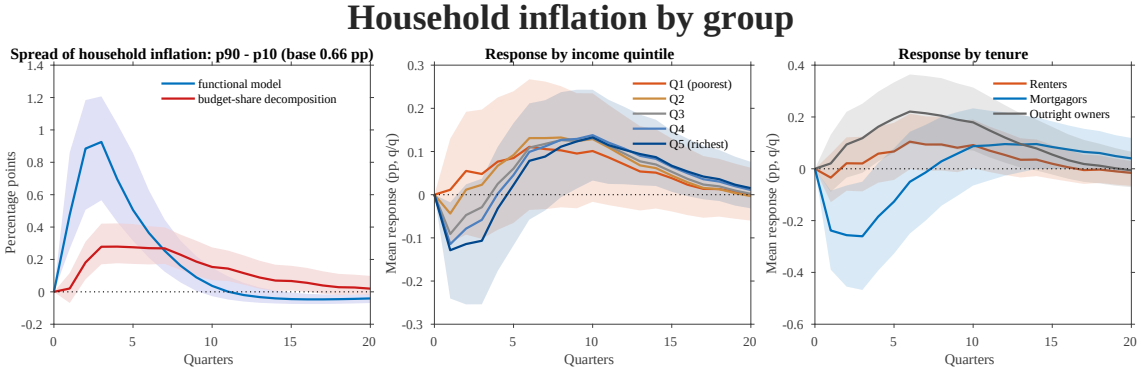


Figure 4: **Household inflation by group.** The budget-share decomposition of the inflation response: each household's spending shares are applied to the responses of the seventeen RPI group price series. Left: the implied change in the p90 − p10 spread of household inflation, against the same statistic from the functional model. Centre and right: mean responses by income quintile and by tenure. Posterior medians with 68% bands; shaded bands in the centre panel are shown for the poorest and the richest quintile only.

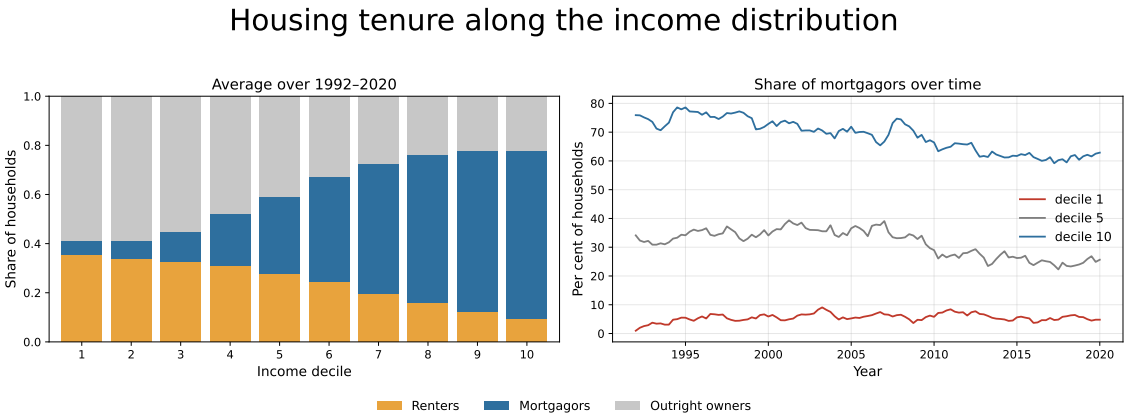


Figure 5: **Housing tenure along the income distribution.** Left: the survey-weighted share of renters, mortgagors and outright owners within each decile of household income, averaged over 1992–2020. Right: the share of mortgagors in the bottom, middle and top decile over time (four-quarter moving average). Tenure is read off the housing split of the survey, which starts in 1992 because the 1991 files record an imputed rent for owner-occupiers rather than their payments: a household with mortgage interest is a mortgagor, one with rent and no mortgage interest a renter, and the remainder are outright owners or live rent-free. This is the gradient behind the income ranking of the inflation responses in Figure 4.

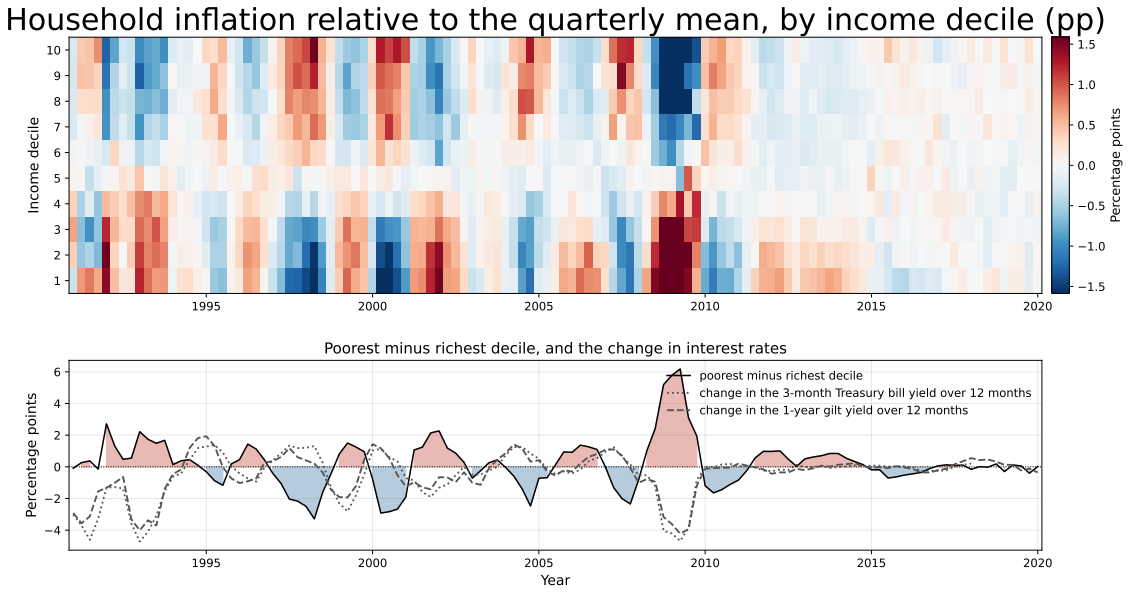


Figure 6: **Household inflation along the income distribution.** Upper panel: the 12-month household RPI inflation rate of each income decile, measured as a deviation from the weighted mean of that quarter, so that red marks deciles facing above-average inflation. Lower panel: the gap between the poorest and the richest decile, with the change in the 3-month Treasury bill yield (dotted) and in the 1-year gilt yield (dashed) over the previous twelve months, all in percentage points. The gap opens when rates fall and closes or reverses when they rise, so that it moves opposite to the rates, most visibly in 2008–09, when the fall in rates lowered the price of mortgage interest for the mortgagors concentrated in the upper deciles. The Treasury bill yield is taken from the macroeconomic panel of Theophilopoulou (2022) and ends in 2018Q1.

falls by about half a point after four years. The compression therefore comes from both ends, but mostly from the bottom moving up towards the middle: after the shock, fewer households are far below the average household. Its size and timing are in line with Theophilopoulou (2022), and so is its location: in her results, the gap between the median and the 10th percentile of income narrows far more than the gap between the 90th percentile and the median. It also resembles the initial fall in the share of low-earning workers that Huber et al. (2025) report for the United States, although their object, the earnings of employed workers, is closer to the employment-income distribution below, which barely moves. The picture is the same after taxes. Appendix E repeats the model for disposable income, gross income minus direct taxes, and finds responses close to those of gross income: direct taxes change the level of income inequality, but hardly its response to the shock.

Figure 7 estimates each income source of (1) as a distribution in its own right, and Figure 19 in the appendix summarises each by its Gini coefficient. The composition argument can therefore be read off the data rather than assumed and it becomes obvious that the four sources behave very heterogeneously.

Employment income, among households that have earnings, barely moves at any horizon. Its Gini response is between -0.1 and -0.2 points, with a band that covers zero throughout: wage dispersion among earners is not the driver. Benefits compress within the first year, by about half a Gini point at the four-quarter horizon, before the response fades in the third year. The density response shows mass leaving the smallest payments, around the 5th percentile of the benefit distribution, and gathering around the 20th: the smallest payments rise relative to the average payment. Benefits make up 84% of the gross income of the bottom income decile but only 2% of the top, as Figure 20 documents. This is therefore the source that sits under the lower tail of the income distribution, where the compression of total income starts.

Investment income and private pensions are the most unequal source in the data, with a baseline Gini of 76, and they compress at every horizon, by 0.6 points after one year and 0.8 after two. As in the previous components, the compression comes from below. Mass leaves the lower half of the source's distribution and gathers between its median and its 95th percentile, while the largest amounts barely move. Figure 20 also shows that this source is not concentrated at the top of the income distribution: it is the largest share of income in the third to fifth deciles, and less than a tenth at the top. Self-employment income moves the other way: its Gini rises by about one point as more households fall far below the average self-employment income, but its bands are wide. The composition channel of Theophilopoulou (2022) is therefore confirmed at the bottom of the distribution, where it runs through the benefit system, but not at the top: neither the wage distribution nor the top of the capital-income distribution falls back.

Household consumption. Consumption compresses earlier and by more than income. Figure 2 shows its dispersion falling faster and reaching its minimum after about two and a half years, while that of income keeps falling for about four years. Figure 18 puts the response of the consumption

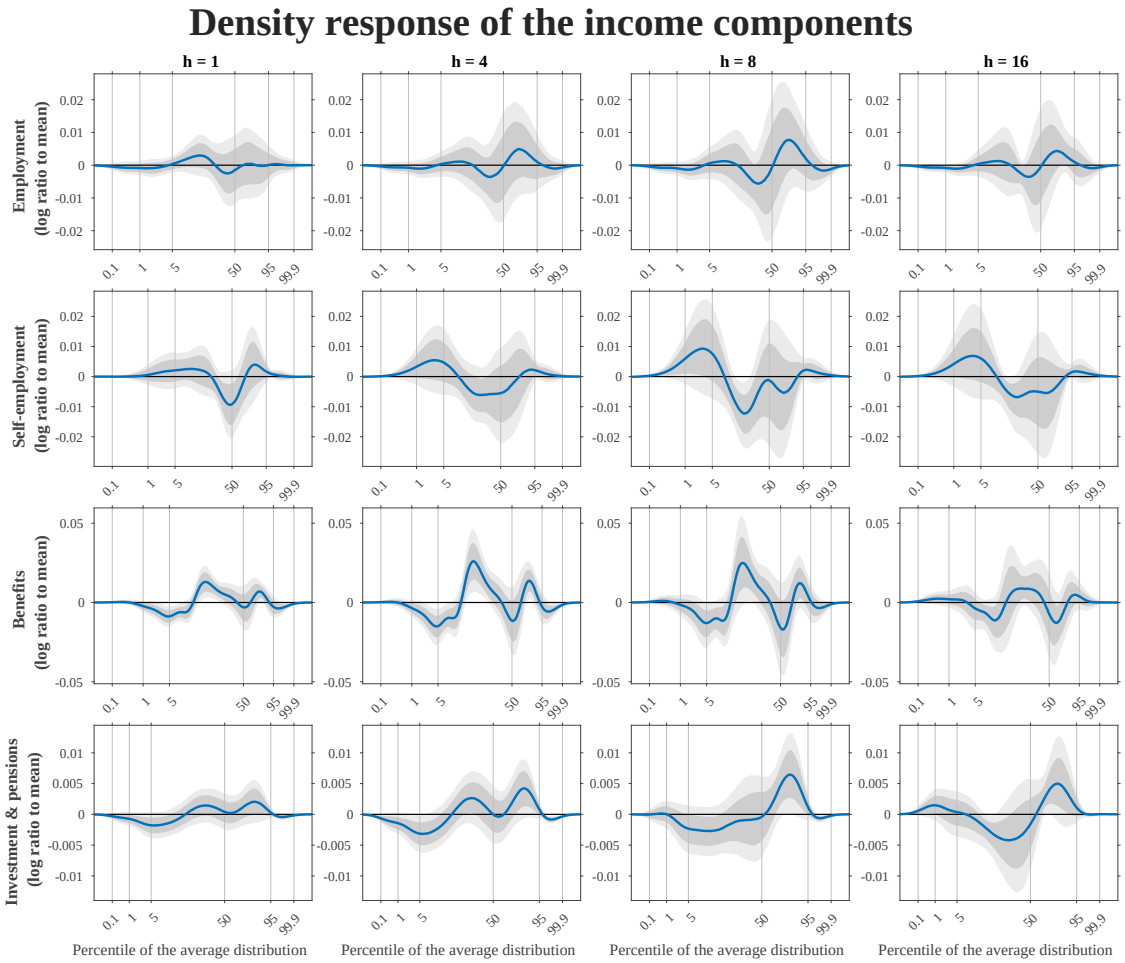


Figure 7: **The components of income.** Functional impulse responses of the four income sources of (1), in the same form as Figure 3: each row is a separate model of one source among the households that receive it, each column a horizon, and the horizontal axis is marked in percentiles of that source's average distribution. The residual category "other income" is not modelled. Posterior medians with 68% and 90% bands.

Gini at more than 0.8 points at the two- to three-year horizon, against about half a point for income after four years. The percentiles in Figure 15 show the same asymmetry as for income: relative to the mean, the 5th percentile gains about five and a half log points and the 95th loses about one and a half. The density response in Figure 3 shows mass leaving the lower tail and, much less, the upper tail, and gathering around the middle. As for income, the compression comes mostly from the bottom moving up.

Figure 8 repeats the exercise for the spending categories of (2), with their Gini coefficients in Figure 19. Two categories compress. Spending on housing and fuel shows the strongest compression of all the objects estimated in this paper, a Gini response of 1.3 points after two years, and its density response drains both tails into the middle, the lower tail much more. Section 5.3 returns to housing costs. Transport compresses at every horizon, by 0.7 to 0.9 Gini points, and again from below: mass leaves the lower part of the distribution and gathers above its middle, while the top

barely moves.

Food behaves differently. Its distribution does not compress: the Gini response is positive, though not distinguishable from zero, and the density response adds mass in the lower tail, consistent with the most constrained households trading down. Alcohol and tobacco show no reliable movement. So, of the two categories that dominate spending at the bottom, housing and fuel compresses and food does not. Figure 21 shows that together they take more than 60% of spending in the bottom decile of the spending distribution and less than 30% in the top one. Along the income distribution the basket is graded in the same way: Figure 22 shows that the bottom income decile spends about 24% of its budget on food against 15% at the top, while the top decile spends about 16% on transport against 8% at the bottom.

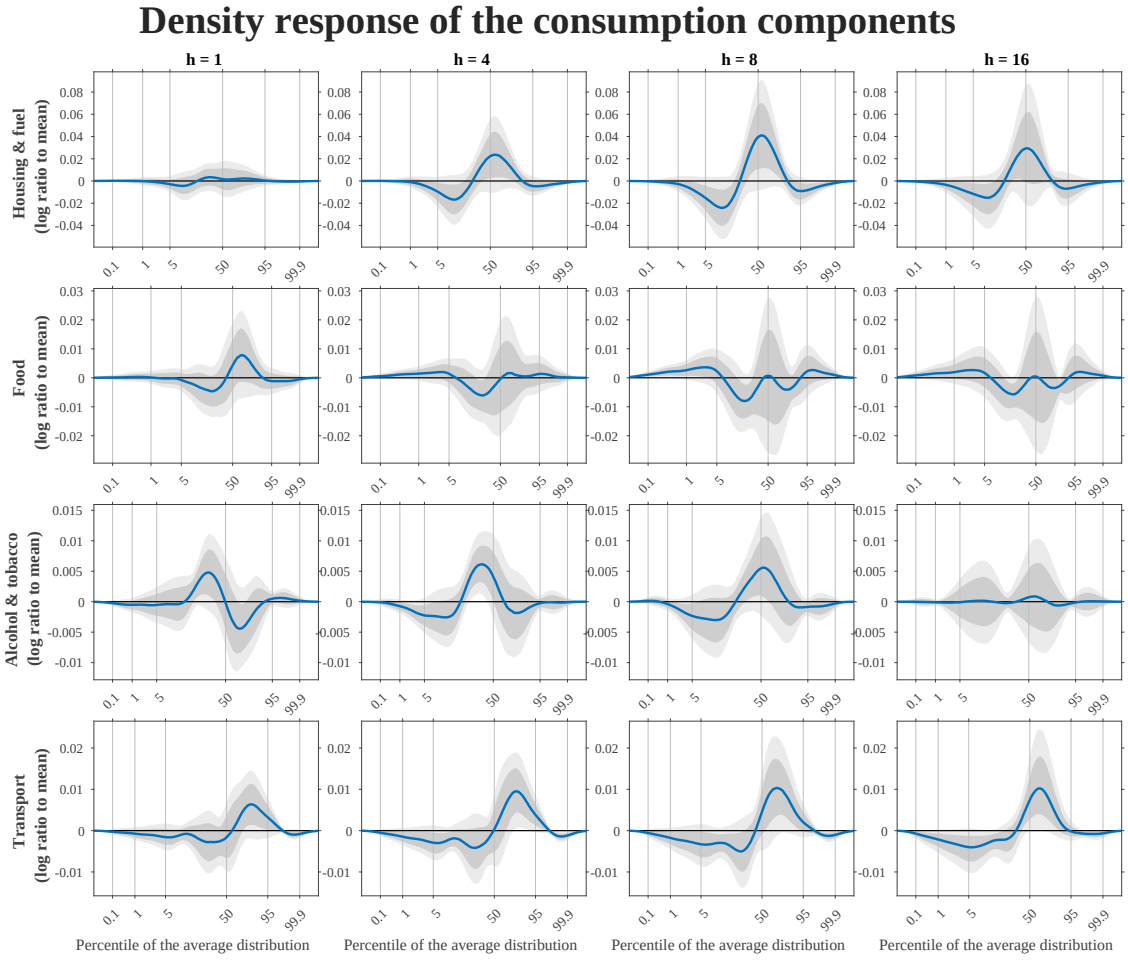


Figure 8: **The components of consumption.** Functional impulse responses of the four spending categories of (2), in the same form as Figure 3: each row is a separate model of one category among the households that report it, each column a horizon, and the horizontal axis is marked in percentiles of that category's average distribution. The residual category "other consumption" is not modelled. Posterior medians with 68% and 90% bands.

Dependence. Since income, consumption and household inflation are observed for the same households, there is reason to believe that a shock could also change how the three distributions

line up within the cross-section, not only narrow each of them. Appendix C.4 carries the copula of the three ranks through a VAR of the same form. The rank correlations barely move. Only the correlation between income and consumption moves by more than its band, and only by -0.003 against a level of 0.47. The shock therefore narrows each distribution without changing how the three rankings line up within a quarter, at least as far as the survey's sample sizes can detect.

5.3 Mortgagors, renters and the cost of housing

The results above show that the cost of living of mortgagors falls after the shock, through the price of mortgage interest, while that of renters barely moves. The same tenure split runs through the literature on how households respond to shocks. Cloyne and Surico (2017) show that households with mortgage debt adjust their consumption strongly to income tax changes, while outright owners do not. Cloyne et al. (2020) show that after a rate cut mortgagors raise their spending markedly, while outright owners do not. And Theophilopoulou (2022) shows that after an uncertainty shock mortgagors cut their consumption relative to renters. The first two measure consumption without housing costs. Because the survey records the housing payments and the inflation rate of every household, three outcomes can be followed for the same groups: the cost of living, what households pay for housing, and what they spend on everything else, as set out in Section 4.3.

Figure 9 shows the responses, all relative to the average household. The first row shows spending. In the left panel, the total spending of renters rises by about 2.5% after two years and that of outright owners by about 2.2% after three, while that of mortgagors falls, by about 1.1% after two years and 2.1% after three. The centre panel splits the spending of mortgagors. The mortgage interest they report falls gradually: by about 4% after two years, with a band that includes zero, and by about 6% after three. The rest of their spending falls by about 1%. The right panel does the same for renters. Their rent payments rise by about 8% after two years, while the rest of their spending dips on impact and barely moves after the first year.

The second row of Figure 9 shows the cost of living. In the left panel, the inflation of mortgagors falls by up to 0.27 percentage points per quarter in the first year, while that of renters rises by up to 0.09 and that of outright owners by up to 0.19. These rates come directly from the households in each group rather than from aggregate prices mapped through baskets and hence confirm the budget-share decomposition of Figure 4. They are larger in magnitude because they are measured relative to all households. The centre panel shows where the fall comes from. The price of mortgage interest falls by up to about 3 percentage points per quarter relative to average inflation, while the rest of the basket of mortgagors rises slightly faster than the average. In the right panel, the price of renting rises by up to 0.2 points per quarter less than average inflation, and the rest of the basket of renters slightly faster.

The third row compares groups. In the left panel, the total spending of mortgagors falls relative to that of renters by about 3.6% after two years and 4.7% after three, the sign that Theophilopoulou

(2022) finds. Without housing payments, the measure of Cloyne and Surico (2017), the gap is much smaller, about 1.3% after two years and 1.5% after three. In the first three quarters it is even positive, because renters cut their other spending on impact. In the right panel, the consumption of the poorest income quintile rises relative to that of the richest by about 5% after two years in total spending and by about 4% without housing payments.

The figure therefore separates three outcomes that are easily treated as one. The first is the cost of living. Household inflation measures how the price of each group's basket changes after the shock, and for mortgagors it falls with the price of mortgage interest. The second is what households pay for housing. Reported payments need not mirror these prices, and here they do not: the mortgage interest that mortgagors report falls by far less than its price, and renters pay more rent although the price of renting falls relative to other prices. Note that a price index for mortgage interest measures the cost of servicing an unchanged mortgage, while reported payments reflect the contracts households actually hold, and who holds them.[9] The third outcome is consumption beyond housing. Outside housing, the prices that different groups face barely differ, so valuing non-housing consumption at each group's own prices leaves both comparisons almost unchanged (dotted lines in the bottom row of Figure 9).

For the mortgage channel, this means that it is clearly at work on the cost of living with weaker effects on spending. The non-housing consumption of mortgagors falls by only about 1.3% relative to that of renters, and most of the gap in total spending comes from renters, whose rent payments rise. The comparison by income, by contrast, does not depend on housing: the consumption of the poorest fifth of households rises relative to that of the richest fifth with and without housing payments and at each group's own prices. The compression seen in Section 5.2 is therefore not an artefact of housing costs or of the price index.

[9] The survey is a repeated cross-section, so group means also move with who belongs to each group. In a VAR of the same form, the share of mortgagors among all households falls by about 1.6 percentage points after three years, and that of renters rises by about 0.7, with a band that only just includes zero. Part of the rise in rent payments may therefore come from households that rent instead of buying (i.e., a form of extensive margin).

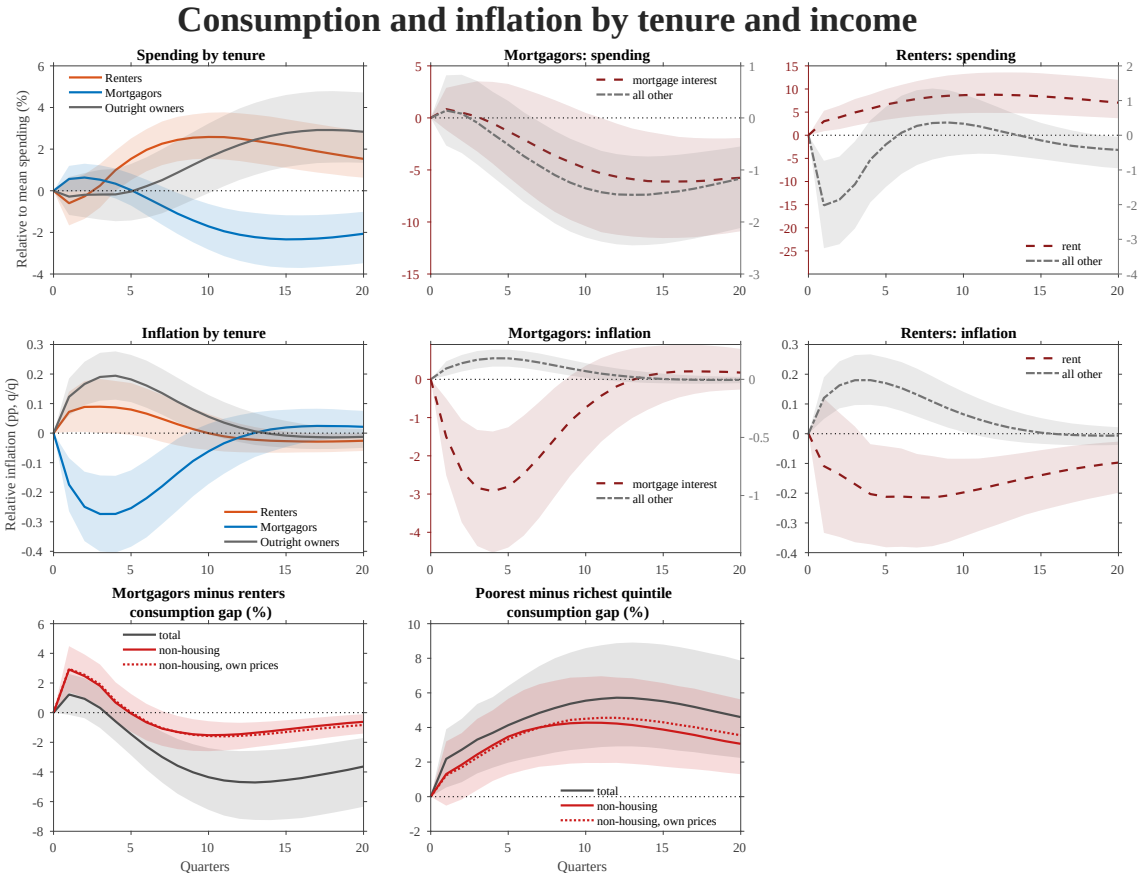


Figure 9: **Consumption and inflation by housing tenure and income.** Responses relative to all households, (13), per capita, 1992Q1–2018Q1, each series from its own VAR. First row: total spending by tenure, and the spending of mortgagors and of renters split into the housing payment that defines the group (mortgage interest, rent) and all other spending, in per cent of the mean spending of all households. Second row: household inflation by tenure, and the inflation of the same parts, in percentage points per quarter relative to the mean inflation of all households. Third row: the consumption gap between mortgagors and renters and between the poorest and the richest income quintile, for total consumption, for non-housing consumption (Cloyne and Surico, 2017), and for non-housing consumption at each group's own prices, (14). Posterior medians with 68% bands; per shock of +0.15 index points on impact.

6 Robustness

Every result in Section 5 rests on a chain of choices: how the shock is identified, how the functional object is truncated, which information enters the VAR, how many lags it has, how top-coded incomes are treated, and which estimator is used. This section changes each choice in turn and reports the same object each time: the response of the cross-sectional standard deviation of the three distributions, the summary statistic of Figure 2. Figure 10 collects the results. The upper row changes the specification, the lower row the estimator. Each variant re-estimates all three models from the microdata; nothing is reweighted after the fact.

Identification. The benchmark orders uncertainty last, so the distribution cannot respond within the quarter. This is the convention of Jurado et al. (2015), and Theophilopoulou (2022)

also uses it as her benchmark. Her robustness section reports the opposite ordering, due to Bloom, in which uncertainty comes first and affects everything on impact. Under that ordering, the inflation response is larger in the first quarters and dies out about four quarters sooner, as an impact response mechanically implies: it lies above the benchmark's 68% band in the first three quarters and below it from the sixth to the thirteenth. The income and consumption responses are indistinguishable from the benchmark from the first quarter onwards. All three now show a small positive effect on impact, which the benchmark ordering rules out by construction. But it is tiny and none of the three is distinguishable from zero: $+0.005$ for income and $+0.003$ for consumption on impact, against troughs of -0.014 and -0.019, with 68% bands that straddle zero. Letting the distributions respond within the quarter therefore adds nothing the data support.

Truncation. The benchmark fixes the number of functional principal components at three for every distribution. This is parsimonious, but each marginal then explains a different share of its own variation, between 87% and 91% (Figure 11). Fixing the *share* instead of the count, at 90% for each marginal, changes only the consumption model, which moves to four components and 92% of the variation. Its standard-deviation response lies on top of the benchmark in Figure 10.

Information set. A VAR with eight variables is a small information set compared with what households and policymakers observe. Theophilopoulou (2022), following Forni and Gambetti (2014), addresses this by adding to the system the first two principal components of a panel of 64 UK macroeconomic and financial series. I do the same with her panel and order the two factors before the macro block.[10] The responses stay within the benchmark's 68% band at every horizon. They keep their sign, shape and timing, and are slightly larger for income at longer horizons.

Lag order. The benchmark uses four lags, following Theophilopoulou (2022). Halving the lag order to two or doubling it to eight keeps the income and consumption responses within the benchmark's 68% band throughout, and the inflation response with eight lags as well. With two lags, the inflation response peaks about a tenth lower, just below the benchmark's band in the third quarter, and decays more slowly: it stays above the benchmark's band from the fourth year on and turns negative only in the last quarter of the five-year horizon.

Top-coded incomes. The baseline replaces censored incomes with a fitted Pareto tail. An alternative used in the literature (e.g. Brewer and Wren-Lewis, 2016) is to trim the tails, excluding observations at the extremes of the income distribution. This involves a clear trade-off: the tails are where the effects of shocks are likely to be most heterogeneous, but also where measurement error is largest. Removing the correction altogether, by taking gross income as recorded and

[10] The panel is quarterly, runs from 1970Q2 to 2018Q1, is complete and is already transformed to stationarity. The series are standardised and the two components are estimated by principal components over the whole panel; together they carry 31% of its variation. The first loads mainly on prices and interest rates and correlates 0.84 with CPI inflation over 1977–2018; the second is a real-activity factor, correlating -0.79 with real GDP growth.

dropping every household flagged as censored (1.05% of the sample, none before 2006), leaves all three responses unchanged in sign, shape and timing. Income compresses somewhat more once the imputed tail is gone, but well within the benchmark's band (Figure 10).

Estimator. The variants above all propagate the shock through a VAR. The last exercise replaces that propagation entirely with the functional local projections of Huber et al. (2025). Each horizon gets its own regression of the loadings h quarters ahead on the uncertainty index, on the contemporaneous values of everything ordered before it, and on four lags of the system. The recursive timing restriction is thus imposed inside the regression rather than through a Cholesky factor (Plagborg-Møller and Wolf, 2021). Estimation follows the Bayesian local projections of Ferreira et al. (2023): the prior is centred on the path implied by the VAR rather than on zero, and its tightness gives equal weight to the projection and to that prior. The estimator therefore nests both approaches: a flat prior returns ordinary local projections, a tight prior returns the VAR. The exercise asks whether the extra flexibility of horizon-by-horizon estimation overturns the benchmark. It does not. The paths are much less smooth and the bands much wider, as is usual for local projections. But for income and consumption the 68% band covers the benchmark at every horizon, and for household inflation at all but four: the second and third quarters, where the projection peaks higher, and two late horizons, including the last, where the projection turns up again as the effective sample shrinks.

Two conclusions follow. First, the results are not an artefact of the ordering, the truncation, the information set, the lag order or the treatment of top incomes. These choices move the responses by less than the width of the posterior band. The exceptions are the larger and shorter-lived inflation response with uncertainty ordered first, and its slower decay with two lags. Second, what the exercises do not pin down is magnitude. The projections put the peak response at 1.4 to 1.7 times the VAR's, with much wider bands, and the specification variants shift the trough for income and consumption by at most a quarter. The one exception is that removing the top-code correction deepens the trough for income by about a third. This is why the paper reads shapes and channels from the functional model and treats magnitudes as approximate throughout.

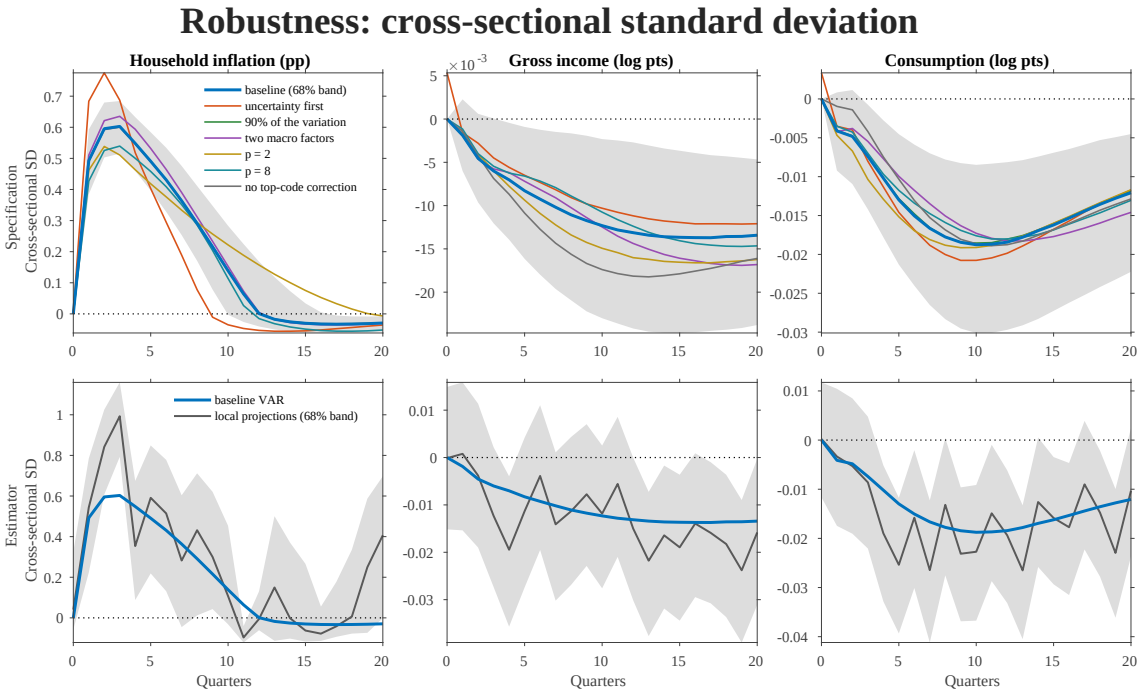


Figure 10: **Robustness.** The response of the cross-sectional standard deviation of each distribution to the same +0.15 uncertainty shock. Upper row: the benchmark (thick line, with its 68% band) against the variations of the specification – uncertainty ordered first, the factor count set by a 90% variance target, two macroeconomic factors added to the information set, two and eight lags in place of four, and gross income as recorded, with every censored household dropped, in place of the Pareto top-code correction. Lower row: the benchmark against the Bayesian local projections of Ferreira et al. (2023), with the projections' own 68% band, over the same five years. Units are percentage points for inflation and log points for income and consumption.

7 Conclusion

This paper asked who bears macroeconomic uncertainty in the United Kingdom. It answered by putting whole cross-sectional distributions, rather than summary statistics, into a VAR with a recursively identified uncertainty shock. Income, consumption and household-specific inflation come from the same households in the same survey. Each income source and each spending category is modelled as a separate distribution, and for housing-tenure groups the cost of living, the housing payments and all other spending are followed side by side.

The answer has two parts that point in opposite directions. In quantities, uncertainty compresses. The distributions of gross income and consumption narrow, by about half a Gini point within four years and by eight tenths of a point within two, and the narrowing comes mainly from the bottom moving up towards the middle, relative to the average household, with a smaller fall at the top. The component models show where this happens. Benefits, the bulk of income at the bottom, compress within the first year; investment and pension income compresses from its lower end; spending on housing and fuel and on transport compresses, while food spending does not; and the wage distribution among earners does not respond at all. This confirms the composition channel of Theophilopoulou (2022) at the bottom of the distribution, where it runs through the

benefit system, but not at the top, and it is now estimated rather than inferred.

In prices, uncertainty does the opposite. The dispersion of the inflation rates households actually face widens for about three years, most of it within the first year and a half. This mainly runs through the price of mortgage interest, which falls together with interest rates after the shock and is paid mostly by households in the upper half of the income distribution.[11] The tenure comparison shows that the cost of living, what households pay and what they consume are distinct outcomes. The lower price of mortgage interest lowers the measured cost of living of mortgagors, and the mortgage interest they report falls by less. Most of the fall in their spending relative to renters comes from the rising rent payments of renters. Beyond housing, the gap is small, and it does not depend on group-specific prices. Through all of this, the dependence between the three rankings does not measurably change: the shock changes the distances within each distribution, not how the distributions line up.

Section 6 asks how much of this depends on how the model is set up. The answer is: very little. Ordering uncertainty first instead of last, as Theophilopoulou (2022) does in her robustness section, changes the early inflation response and little else. Choosing the number of functional principal components by a fixed variance share moves the consumption model to four factors and leaves its response unchanged. Adding the first two principal components of the 64-series panel of Theophilopoulou (2022), the informational-sufficiency correction of Forni and Gambetti (2014), leaves all three responses in place. Halving or doubling the lag order does the same, except that with two lags the inflation response decays more slowly. Bayesian local projections, which give up efficiency for robustness to dynamic misspecification, produce much noisier paths, but their bands cover the benchmark at nearly every horizon and their shape is comparable. Removing the Pareto top-code correction, and with it every censored household, leaves the picture untouched.

Two implications follow. For measurement, an aggregate price index is the index of nobody in particular, and the gap is not a curiosity. It moves systematically with interest rates, as the raw data in Figure 6 show, and after an uncertainty shock it moves in favour of richer households in the first year, in the opposite direction to the compression of income and spending. Distributional work that uses one deflator for all households misses this by construction. Comparisons of housing-tenure groups, in turn, depend on whether housing payments count as consumption, and the price of housing need not move like what households pay for it (Section 5.3). For policy, the distributional footprint of an uncertainty shock depends on the composition of household income, the shopping basket and housing tenure, rather than on the wage distribution. The same shock narrows the gaps in income and spending, through the transfers and necessities that dominate the bottom, and widens the gaps in the cost of living, through the mortgages held towards the top.

[11] The decline in rates is part of the estimated response to uncertainty, not an independent policy easing.

References

Bayer, C., L. Calderon, and M. Kuhn (2026). Distributional dynamics. CEPR Discussion Paper 19829, Centre for Economic Policy Research.

Beckmann, J. and R. L. Czudaj (2026). Uncertainty shocks and inflation: The role of credibility and expectation anchoring. *Journal of International Money and Finance 160*, 103472.

Bilbiie, F. O., S. M. Galaasen, R. S. Gürkaynak, M. Mæhlum, and K. Molnar (2025). HANKSSON. CEPR Discussion Paper 20090, Centre for Economic Policy Research.

Binder, C. C. (2017). Economic policy uncertainty and household inflation uncertainty. *The B.E. Journal of Macroeconomics 17*(2).

Bjørnland, H. C., Y. Chang, and J. L. Cross (2026). Oil and the stock market revisited: A mixed functional VAR approach. *Quantitative Economics 17*(2), 541–589.

Bloom, N. (2009). The impact of uncertainty shocks. *Econometrica 77*(3), 623–685.

Brewer, M. and L. Wren-Lewis (2016). Accounting for changes in income inequality: Decomposition analyses for the UK, 1978–2008. *Oxford Bulletin of Economics and Statistics 78*(3), 289–322.

Chan, J. C. C. (2022). Asymmetric conjugate priors for large Bayesian VARs. *Quantitative Economics 13*(3), 1145–1169.

Chang, M., X. Chen, and F. Schorfheide (2024). Heterogeneity and aggregate fluctuations. *Journal of Political Economy 132*(12), 4021–4067.

Chang, M. and F. Schorfheide (2024). On the effects of monetary policy shocks on income and consumption heterogeneity. NBER Working Paper 32166, National Bureau of Economic Research.

Chang, Y., F. Gómez-Rodríguez, and G. H. Hong (2022). The effects of economic shocks on heterogeneous inflation expectations. IMF Working Paper WP/22/132, International Monetary Fund.

Chang, Y., S. Kim, and J. Y. Park (2025). How do macroaggregates and income distribution interact dynamically? a novel structural mixed autoregression with aggregate and functional variables. CAMA Working Paper 7/2025, Centre for Applied Macroeconomic Analysis.

Choi, S. and J. Phi (2026). Impact of uncertainty shocks on income and wealth inequality. *Journal of Money, Credit and Banking 58*(3), 821–852.

Cloyne, J., C. Ferreira, and P. Surico (2020). Monetary policy when households have debt: New evidence on the transmission mechanism. *The Review of Economic Studies 87*(1), 102–129.

Cloyne, J. S. and P. Surico (2017). Household debt and the dynamic effects of income tax changes. *The Review of Economic Studies 84*(1), 45–81.

Crawford, I. and Z. Smith (2002). Distributional aspects of inflation. IFS Commentary 90, Institute for Fiscal Studies, London.

Deheuvels, P. (1979). La fonction de dépendance empirique et ses propriétés: un test non paramétrique d'indépendance. *Académie Royale de Belgique, Bulletin de la Classe des Sciences 65*, 274–292.

Ettmeier, S. (2024). No taxation without reallocation: The distributional effects of tax changes. Working paper, University of Bonn.

Ferreira, L. N., S. Miranda-Agrippino, and G. Ricco (2023). Bayesian local projections. *The Review of Economics and Statistics*.

Fischer, M. M., F. Huber, and M. Pfarrhofer (2021). The regional transmission of uncertainty shocks on income inequality in the united states. *Journal of Economic Behavior & Organization 183*, 887–900.

Forni, M. and L. Gambetti (2014). Sufficient information in structural VARs. *Journal of Monetary Economics 66*, 124–136.

Huber, F., M. Marcellino, and T. Tornese (2025). The distributional effects of economic uncertainty. CEPR Discussion Paper 20050, Centre for Economic Policy Research.

Inoue, A. and B. Rossi (2021). A new approach to measuring economic policy shocks, with an application to conventional and unconventional monetary policy. *Quantitative Economics 12*(4), 1085–1138.

Istrefi, K. and A. Piloiu (2014). Economic policy uncertainty and inflation expectations. Working Paper 511, Banque de France.

Jurado, K., S. C. Ludvigson, and S. Ng (2015). Measuring uncertainty. *American Economic Review 105*(3), 1177–1216.

Koop, G., S. McIntyre, J. Mitchell, and P. Wu (2026). Incorporating micro data into macro models using pseudo VARs. Working Paper 26-04, Federal Reserve Bank of Cleveland.

Lauper, C. and G. Mangiante (2025). Monetary policy shocks and inflation inequality. *International Journal of Central Banking 21*(4), 135–190.

Lenza, M. and J. Slacalek (2024). How does monetary policy affect income and wealth inequality? evidence from quantitative easing in the euro area. *Journal of Applied Econometrics 39*(5), 746–765.

Marcellino, M., A. Renzetti, and T. Tornese (2024). Firm heterogeneity and macroeconomic fluctuations: A functional VAR model. Working Paper 2411.05695, arXiv.

Plagborg-Møller, M. and C. K. Wolf (2021). Local projections and VARs estimate the same impulse responses. *Econometrica 89*(2), 955–980.

Sancetta, A. and S. Satchell (2004). The Bernstein copula and its applications to modeling and approximations of multivariate distributions. *Econometric Theory 20*(3), 535–562.

Sklar, A. (1959). Fonctions de répartition à n dimensions et leurs marges. *Publications de l'Institut de Statistique de l'Université de Paris 8*, 229–231.

Theophilopoulou, A. (2022). The impact of macroeconomic uncertainty on inequality: An empirical study for the united kingdom. *Journal of Money, Credit and Banking 54*(4), 859–884.

A Top-coded incomes

From 2006 onwards, the survey releases gross weekly household income in top-coded form: every household above a disclosure threshold is recorded at one and the same amount. The censoring is therefore visible as a mass point at the maximum of each year's income distribution. I treat a year as top-coded when at least five households, and at least 0.5% of its sample, share that maximum; no year before 2006 meets this condition. Between 2006Q1 and 2020Q1 the censoring point rises from £1,644 to £2,496 a week, and between about 3% and 5% of the households in each quarter are censored.

I treat the censored tail as Chang et al. (2024) treat top-coded earnings, in two respects. First, the censoring point and the number of censored households are taken as known, and the object of interest is the distribution of uncensored income. A censored household tells us only that its income lies at or above the censoring point; the recorded amount itself is the same for all such households and carries no further information. Second, the density above the censoring point is extrapolated with a Pareto tail. In Chang et al. (2024), this is the role of the linear rightmost segment of their log-spline density. They impose it because a cubic segment can make the estimated density increase between the top-coded value and the upper bound of the support; and a log density that is linear in log income is a Pareto density in levels. Where they build the censoring into the likelihood of the log-spline density, I impute the censored incomes from the fitted tail. This way, the kernel densities of Appendix C.1 and the copula of Appendix C.4 can be computed from household-level data in every quarter.

The estimator. Within quarter t, let $y_{h,t}$ denote the gross weekly income of household h before equivalisation, $\bar{y}_t$ the censoring point, $\mathcal{C}_t$ the set of censored households, $\tau_{h,t} \in (0,1)$ the survey-weighted mid-rank of $y_{h,t}$, and $\tau_t^c = \min_{h \in \mathcal{C}_t} \tau_{h,t}$ the rank at which the censoring begins. A Pareto tail above the censoring point, $1 - F_t(y) \propto y^{-\alpha_t}$ for $y \geq \bar{y}_t$, has the quantile function

$$Q_t(\tau) = \bar{y}_t \left(\frac{1 - \tau_t^c}{1 - \tau} \right)^{1/\alpha_t}, \qquad \tau \geq \tau_t^c, \tag{15}$$

so that $\log Q_t(\tau) - \log \bar{y}_t$ is linear in $\log(1 - \tau_t^c) - \log(1 - \tau)$, with slope $1/\alpha_t$. The slope is estimated by survey-weighted least squares, without an intercept, on the uncensored households in the tenth of the distribution immediately below the censoring point, $\tau_t^c - 0.1 \leq \tau_{h,t} < \tau_t^c$, about 130 households per quarter. Leaving out the intercept anchors the fitted quantile function at the censoring point, so that it is continuous there. Each censored household is then assigned the fitted quantile at its own rank,

$$\tilde{y}_{h,t} = \min \left\{ Q_t(\tau_{h,t}),\ 35\, \bar{y}_t \right\}, \qquad h \in \mathcal{C}_t. \tag{16}$$

Across the censored quarters the estimated tail index α_t lies between 2.2 and 4.2, with a median of 2.8.

Adjustments for the U.K. data. Three details are specific to the survey. First, the treatment is applied to household income before equivalisation, because that is where the top-code is imposed. Dividing by the square root of household size would spread the single censoring point over many amounts and make it impossible to detect, so the equivalence scale of Section 3 is applied to the imputed incomes afterwards. Second, the censored households share one recorded amount, so it is not identified which of them receives which quantile in (16). Their ranks are therefore ordered at random, with a fixed seed, which keeps the ranks uniform. Third, a Pareto quantile function is unbounded as $\tau \to 1$, so the imputed incomes are capped at 35 times the censoring point. This is just above the largest ratio of the maximum to the 96th percentile in the uncensored quarters of 1996–2005, 33.7; the cap binds in 2 of the 57 censored quarters. The same procedure is applied to disposable income (Appendix E).

The tail above the censoring point remains an extrapolation, not data. Section 6 therefore also estimates the three baseline models on gross income as recorded, with every censored household dropped instead. The responses are unchanged in sign, shape and timing.

B From survey spending codes to RPI price indices

Table 1 documents the construction of the household inflation rates of Section 3: which survey spending codes are pooled to form each weight $s_{h,g,t}$ in (3), which published RPI index prices it, and over which period each correspondence applies. The mapping is constant for the thirteen commodity groups and changes only for housing, where the survey's own components are used from 1992 onwards.

From published indices to a household's inflation rate. Each household's inflation rate is built in four steps. Every number that enters it is either a published RPI index or an amount recorded in the survey.

Step 1: the price change of each group. For each RPI index in the last column of Table 1, the monthly values published by the ONS (series MM23, January 1987 = 100) are averaged to quarters, giving $\overline{\text{RPI}}_{g,t}$ for group g in quarter t. The quarter-on-quarter and 12-month inflation rates of the group are

$$\pi_{g,t} = 100\left(\frac{\overline{\text{RPI}}_{g,t}}{\overline{\text{RPI}}_{g,t-1}} - 1\right), \qquad \pi^{(12)}_{g,t} = 100\left(\frac{\overline{\text{RPI}}_{g,t}}{\overline{\text{RPI}}_{g,t-4}} - 1\right). \tag{17}$$

One price is not published as a single index: for repairs, maintenance and insurance, $\pi_{g,t}$ is the average of the quarter-on-quarter changes of the three RPI sections in the last row of the table, weighted by their published RPI weights of that year, and the corresponding index is chained from these changes. Miscellaneous spending, for which the RPI has no group, is priced with the all-items index.

Step 2: spending on each group. For household h, interviewed in quarter t, $E_{h,g,t}$ is its weekly spending on group g, read from the survey codes in the middle column of the table. The priced groups $\mathcal{G}_t$ are the thirteen commodity groups of Panel A plus housing from Panel B: in 1991, net housing as a single group; from 1992, its five components (rent, mortgage interest, council tax and rates, water charges, and repairs, maintenance and insurance). The five components are built so that they add up exactly to net housing (p536). For the few households whose components do not reconcile, housing stays a single group priced with the combined housing index.

Step 3: spending shares. The household's total spending is the sum over all priced groups, $E_{h,t} = \sum_{g\in\mathcal{G}_t} E_{h,g,t}$. Because the groups exhaust the survey's total expenditure (p550), $E_{h,t}$ is the household's total expenditure, plus the council tax added back in 1998Q2–1999Q1 (see the notes to the table). The spending shares are

$$s_{h,g,t} = \frac{E_{h,g,t}}{E_{h,t}}, \qquad \sum_{g\in\mathcal{G}_t} s_{h,g,t} = 1. \tag{18}$$

A share can be negative when net spending on a group is negative. Motoring expenditure, for example, is recorded net of vehicle sales, so a household that sold a car during its diary period can

have a negative motoring share; the shares still sum to one.

Step 4: the household's inflation rate. Combining Steps 1 and 3 gives (3),

$$\pi_{h,t} = \sum_{g \in \mathcal{G}_t} s_{h,g,t}\, \pi_{g,t}, \qquad \pi^{(12)}_{h,t} = \sum_{g \in \mathcal{G}_t} s_{h,g,t}\, \pi^{(12)}_{g,t}.$$

Each household is observed in one quarter only, so its rate applies the basket it buys in that quarter to the price changes between the previous quarter and the quarter of its interview (for the 12-month rate, over the previous four quarters). Households with zero or negative total spending have no inflation rate. The models of Section 4 use the quarter-on-quarter rate $\pi_{h,t}$, expressed relative to the survey-weighted mean of its quarter and trimmed at the 0.5th and 99.5th percentiles (Section 3); the figures of Section 3 use the 12-month rate $\pi^{(12)}_{h,t}$. As a check, the mean of $\pi^{(12)}_{h,t}$ weighted by survey weight times total spending tracks the published all-items RPI (Figure 1, right panel).

How the groups relate to the five consumption categories. The five categories of (2) are coarser sums of the same spending. They are used only for the consumption-component models; household inflation is always computed at the level of the groups in Table 1. Housing and fuel is net housing plus fuel, light and power (p536 + p537). Alcohol and tobacco matches alcoholic drink plus tobacco (p539 + p540) for at least 99.5% of households, and transport matches motoring plus fares and other travel (p545 + p546) for at least 94%. Food is the survey's own food total, which equals the food group p538 up to 1994–95 and from 1995–96 is on average about £1 a week below it. Other consumption is total expenditure minus these four categories: clothing and footwear, household goods, household services, personal goods and services, leisure goods, leisure services and miscellaneous spending, plus the small differences just described. Splitting each category of (2) into these groups, and splitting housing as in (4), gives exactly the groups on which (3) works.

Table 1: Survey spending codes and the RPI indices that price them

Period	Survey codes pooled	RPI price index (CDID)
Panel A: commodity groups		
1991Q1–2020Q1	p537 fuel, light and power	Fuel and light (CHBG)
	p538 food	Food and catering (CHBS)
	p539 alcoholic drink	Alcoholic drink (CHBD)
	p540 tobacco	Cigarettes and tobacco (CHBE)
	p541 clothing and footwear	Clothing and footwear (CHBJ)
	p542 household goods	Household goods (CHBH)
	p543 household services	Household services (CHBI)
	p544 personal goods and services	Personal goods and services (CHBQ)
	p545 motoring expenditure	Motoring expenditure (CHBK)
	p546 fares and other travel	Fares and other travel (CHBR)
	p547 leisure goods	Leisure goods (CHBL)
	p548 leisure services	Leisure services (CHBM)
	p549 miscellaneous	All items (CHAW)
Panel B: housing		
1991Q1–1991Q4	p536 net housing, priced as one group	Housing (CHBF)
1992Q1–2020Q1	b010 + b020 net rent	Rent (DOBP)
	b130 + b150 mortgage interest	Mortgage interest (DOBQ)
	b050 water charges	Water and other charges (DOBS)
	p281 − (b010 + b020) − (b130 + b150) − b050 − b060, the residual of the survey's housing subtotal	Council tax and rates (DOBR)
	p536 − p281 + b060 repairs, maintenance and insurance	Repairs and maintenance (DOBT), DIY materials (DOBU) and dwelling insurance (DOBV), combined with RPI weights (CZXH, CZXI, CZXJ)

Notes. Prices are the published RPI group and section indices (ONS series MM23, January 1987 = 100), averaged to quarters. Survey codes are given in their canonical form: the same variable carries an `x` prefix up to the 1997–98 files, appears unprefixed from 1998–99 to 2009 and in upper case from 2010, with `t` or `tp` suffixes marking the totals that include children and the anonymised versions; each code is resolved to the spelling of the file it is read from; in the 1994–95 to 1996–97 files the two rent codes carry a `p` suffix. The thirteen groups of Panel A plus net housing (p536) exhaust total household expenditure (p550) in every year, so the shares $s_{h,g,t}$ sum to one by construction. Three further conventions apply. In 1991 the survey records an imputed rent for owner-occupiers rather than their payments, so housing is not split in that year. In 1998Q2–1999Q1 council tax is recorded in p281 but omitted from p536, and hence from total expenditure, for about 47% of households; the recorded amount is added back to those households' baskets. Finally, households whose five housing components do not reconcile with p536, fewer than 0.5% in any year, keep their housing spending on the combined housing index (CHBF).

C Methodological details

C.1 The density estimator and its factors

The estimator. Within each quarter, household h enters with its transformed value $r_{h,t}$ – the log deviation of its income or consumption from the quarter's weighted mean, or its inflation rate relative to the quarter's weighted mean (Section 3) – and with its survey weight, normalised to $\omega_{h,t}$ so that $\sum_h \omega_{h,t} = 1$. Following BCC, the density of quarter t is the Gaussian kernel estimate

$$g_t(r) = \sum_{h=1}^{n_t} \omega_{h,t} \, \frac{1}{\zeta_t} \, \varphi\!\left(\frac{r - r_{h,t}}{\zeta_t}\right), \qquad \zeta_t = 1.06 \, \hat{\varsigma}_t \, n_t^{-1/5}, \tag{19}$$

where φ is the standard normal density, n_t the number of households in the quarter and $\hat{\varsigma}_t$ the survey-weighted standard deviation of the $r_{h,t}$. The bandwidth ζ_t is Silverman's rule of thumb, the choice of BCC, and the survey weights are the one change their estimator needs for household

data: a survey is representative of the population only once it is weighted. The estimator is a density by construction, non-negative and integrating to one, and it plays for the marginals the role that the Bernstein copula of Appendix C.4 plays for the dependence.

The grid. Each density is evaluated on an equispaced grid $r_1 < \cdots < r_M$, so that quarter t is represented by the vector $(g_t(r_1), \ldots, g_t(r_M))'$. The grid spans the pooled range of the trimmed data, extended on each side by three bandwidths of the quarter with the widest kernel, so that every quarter's density lies on it. The spacing is then chosen so that at least ten grid points fall within the bandwidth of the narrowest quarter: M is the smallest power of two that satisfies this condition, and never below the 1,024 points of BCC. The grid is thus common to all quarters of a density series, while its size may differ across series; M lies between 1,024 and 8,192 here.

From densities to factors. The densities are demeaned over time as in (5), $f_t = g_t - \bar{g}$, so that each f_t integrates to zero. Collecting the vectors f_t' in the $T \times M$ matrix $\mathbf{F}$, the FPCs are the eigenvectors of the sample covariance matrix,

$$\hat{\Gamma}\, v_k = \lambda_k v_k, \qquad \hat{\Gamma} = \frac{1}{T}\,\mathbf{F}'\mathbf{F}, \tag{20}$$

ordered by their eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots$: on the grid, FPCA is ordinary principal component analysis. Each v_k is a curve over the support, positive where the component adds households and negative where it removes them, and, as a linear combination of the f_t, it integrates to zero, so that a component can move households along the support but cannot change their number. The factors are the projections

$$c_{kt} = f_t' v_k, \qquad k = 1, \ldots, m, \tag{21}$$

the grid counterpart of (7). They are taken net of quarter dummies, because the survey is not seasonally adjusted, and enter the VAR (9) next to the five macroeconomic variables.

C.2 How many functional principal components

Figure 11 shows the share of the variation of each demeaned density that each FPC explains, and the cumulative share, which is the basis for setting $m = 3$ in Section 4.1.

C.3 The prior

Following Chan (2022), the VAR (9) is written in structural form,

$$AZ_t = \sum_{i=1}^{p} B_i Z_{t-i} + e_t, \qquad e_t \sim \mathcal{N}\big(0, \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)\big)\,, \tag{22}$$

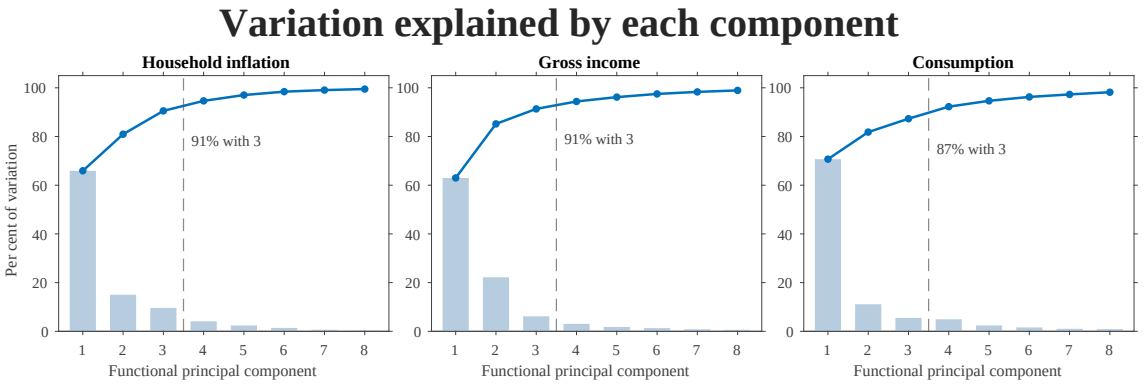


Figure 11: **How many functional principal components.** For each marginal, the share of the variation of the demeaned densities explained by each FPC (bars) and cumulatively (line), with the dashed line marking the $m = 3$ used throughout. The first component alone carries 63–71% of the variation and the first three carry 87–91%; the fourth adds three to five percentage points and the remainder is close to flat.

where A is lower triangular with ones on its diagonal, so that $\tilde{A}_i = A^{-1}B_i$ and $\Sigma = A^{-1}\,\mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)A^{-1\prime}$. The variables are ordered as in the Cholesky identification of Section 4.2, so A carries the same recursive structure. Every variable is demeaned before estimation, so the model has no constant, as in BCC. Equation j of (22) is therefore a linear regression of $Z_{j,t}$ on p lags of all n variables and on the contemporaneous values of the $j-1$ variables ordered before it, with coefficient vector θ_j and error variance σ_j^2.

The asymmetric conjugate prior. The prior is normal-inverse-gamma equation by equation,

$$\theta_j \mid \sigma_j^2 \sim \mathcal{N}\big(m_j,\ \sigma_j^2 V_j\big)\,, \qquad \sigma_j^2 \sim \mathcal{IG}(\nu_j, S_j), \tag{23}$$

independently across equations. Because the equations are a priori independent and each prior is conjugate for its own regression, the posterior factors into n normal-inverse-gamma posteriors that are sampled directly, without a Gibbs sampler. The natural conjugate prior achieves the same only by giving the reduced-form coefficients a prior covariance of Kronecker form, which forces the same shrinkage on a variable's own lags as on the lags of every other variable; the asymmetric prior removes that restriction. The prior mean m_j is zero on every coefficient, the natural centre here because every variable is demeaned and the loadings are stationary by construction. V_j is diagonal, with elements

$$\frac{\kappa_1}{l^2 s_j^2}\ \ (\text{own lag } l), \qquad \frac{\kappa_2}{l^2 s_k^2}\ \ (\text{lag } l \text{ of variable } k \neq j), \qquad \frac{\kappa_3}{s_k^2}\ \ (\text{contemporaneous } Z_{k,t},\ k < j), \tag{24}$$

where s_k^2 is the residual variance of a univariate AR(4) for variable k. Since the prior covariance

of θ_j is scaled by σ_j^2, the implied prior variance of an own lag is about κ_1/l^2 and that of lag l of another variable k about $(\kappa_2/l^2)\,\sigma_j^2/s_k^2$: the familiar Minnesota form, with the variance falling in the lag and the ratio of variances correcting for the units of the variables. I use the values of Chan's replication code: $\kappa_1 = 0.04$ and $\kappa_2 = \kappa_1^2 = 0.0016$, so that the lags of other variables are shrunk towards zero far more strongly than own lags, $\kappa_3 = 1$, $\nu_j = 1 + j/2$ and $S_j = s_j^2/2$. As in that code, the prior variances of the lag coefficients are set for the reduced form and translated into the implied prior on the structural coefficients.

Sampling. Each pair (θ_j, σ_j^2) is drawn directly from its normal-inverse-gamma posterior, 5,000 times; since the draws are independent, no burn-in is needed. A draw whose companion matrix has an eigenvalue on or outside the unit circle is redrawn, up to 1,000 times. Each retained draw is mapped into the reduced form, identified with the Cholesky factor of its own Σ as in (10), and turned into impulse responses, so that every band in the paper is a posterior band of the reported statistic itself.

C.4 The dependence structure

The marginals show how mass moves within each distribution, but not where the affected households sit in the other two: the household with the highest income need not have the highest consumption or face the highest inflation. Because the three variables are observed for the same households, their joint density splits into the marginals and a copula (Sklar, 1959):

$$f_t(y, c, \pi) = \underbrace{f_t^Y(y)\, f_t^C(c)\, f_t^\Pi(\pi)}_{\text{marginals}} \times \underbrace{c_t\big(F_t^Y(y),\, F_t^C(c),\, F_t^\Pi(\pi)\big)}_{\text{dependence}}, \tag{25}$$

where F_t^Y, F_t^C, F_t^Π are the marginal distribution functions of income, consumption and household inflation. The copula density c_t is the joint density of the three ranks. It is unchanged by any increasing transformation of the variables, has uniform margins, and equals one everywhere under independence. Carried through the VAR, it asks whether the shock changes how the three distributions line up within the cross-section, a question the marginal models cannot answer. Marcellino et al. (2024) instead approximate a joint distribution directly with tensor methods; the copula keeps the dependence separate from the marginals.

The estimator. Within each quarter, household h is assigned its survey-weighted ranks $(u_{h,t}, v_{h,t}, w_{h,t}) \in [0,1]^3$ in income, consumption and household inflation, and $\hat{p}_{jkl}^t$ is the weighted share of households in rank cell (j,k,l) of a q^3 grid, the empirical copula (Deheuvels, 1979). Since the marginals are smoothed (Appendix C.1), the copula is smoothed too, with the Bernstein copula density (Sancetta

and Satchell, 2004),

$$\hat{c}_t(u,v,w) = q^3 \sum_{j,k,l=1}^{q} \hat{p}^t_{jkl}\; b_{j,q}(u)\, b_{k,q}(v)\, b_{l,q}(w), \qquad b_{j,q}(u) = \binom{q-1}{j-1} u^{\,j-1}(1-u)^{\,q-j}, \tag{26}$$

where the degree q plays the role of a bandwidth, set to $q = 12 \approx n_t^{1/3}$ for about 1,600 households per quarter. The estimator is a copula by construction, and it is evaluated as its average over each cell of a $4 \times 4 \times 4$ grid of rank quartiles.[12]

Factors and VAR. The 64 cell averages of each quarter form a vector that is treated exactly like a marginal density on its grid: it is demeaned over time and decomposed into principal components as in (20). Every component sums to zero along each of its three directions. It can therefore move households between rank cells but cannot change the margins, and the reconstructed copula keeps uniform margins exactly. The three leading factors, net of quarter dummies, take the place of the loadings of a marginal in (9), with the same macro block, prior and identification and the sample of the household inflation model. The responses are mapped back into the cells as in (11) and added to the copula of the last quarter as in (12), draw by draw.

Results. Figure 12 shows the responses of the three rank correlations. Income and consumption ranks move closely together, with a rank correlation of 0.47 in the estimated copula, while household inflation is almost unrelated to either, with rank correlations of 0.04.[13] The income–inflation correlation rises by about 0.025 at the one- to two-year horizon and the consumption–inflation correlation by about 0.035, but neither has a 68% band that excludes zero. The income–consumption correlation declines from the second year on with a band that excludes zero, but only by -0.003 against a level of 0.47: statistically visible, economically negligible. Figure 13 shows the same picture cell by cell: in the two panels that involve inflation, mass moves towards the diagonal, but only in some cells do the bands exclude zero (indicated by the dot inside a cell).

The shock therefore narrows distances without changing how the three rankings line up within a quarter. This fits the view that ranks are tied to slow-moving characteristics such as age, education, occupation and tenure. Even the mortgage channel, which does shift mortgagors within the inflation distribution, is not recovered precisely. The copula lets the dependence move freely, without the structure of the budget-share decomposition or of the tenure comparison of Section 5.3, and therefore contains less information about any particular channel.

[12] Re-estimating the copula model with the unsmoothed chequerboard copula gives the same results: the same signs, the same horizons and the same verdicts, with the density-response panels correlating at 0.98 across the two estimators. Those estimates are available on request.

[13] These are the rank correlations of the copula as estimated, smoothed and read off the grid of 4^3 cells; both steps pull a correlation towards zero, so the income–consumption value is lower than the rank correlation computed household by household.

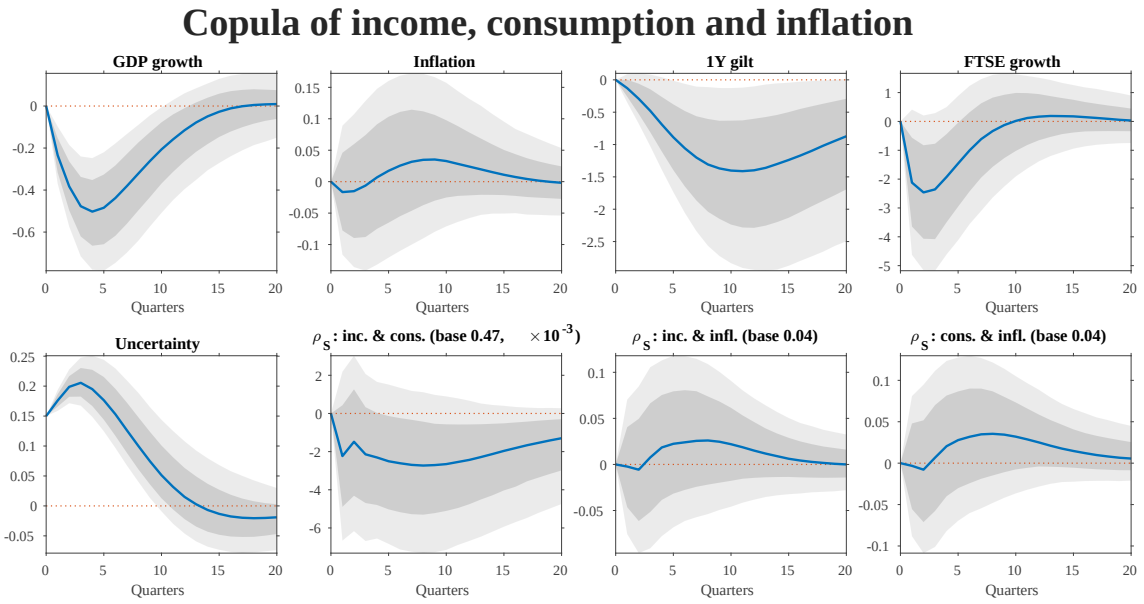


Figure 12: **The copula model.** The macroeconomic block of the trivariate copula model and the responses of the three pairwise Spearman rank correlations, ρ_S. The panel titles give the baseline level of each correlation in the estimated copula. Note the scale of the income–consumption panel, which is in units of 10^{-3}. Posterior medians with 68% and 90% bands.

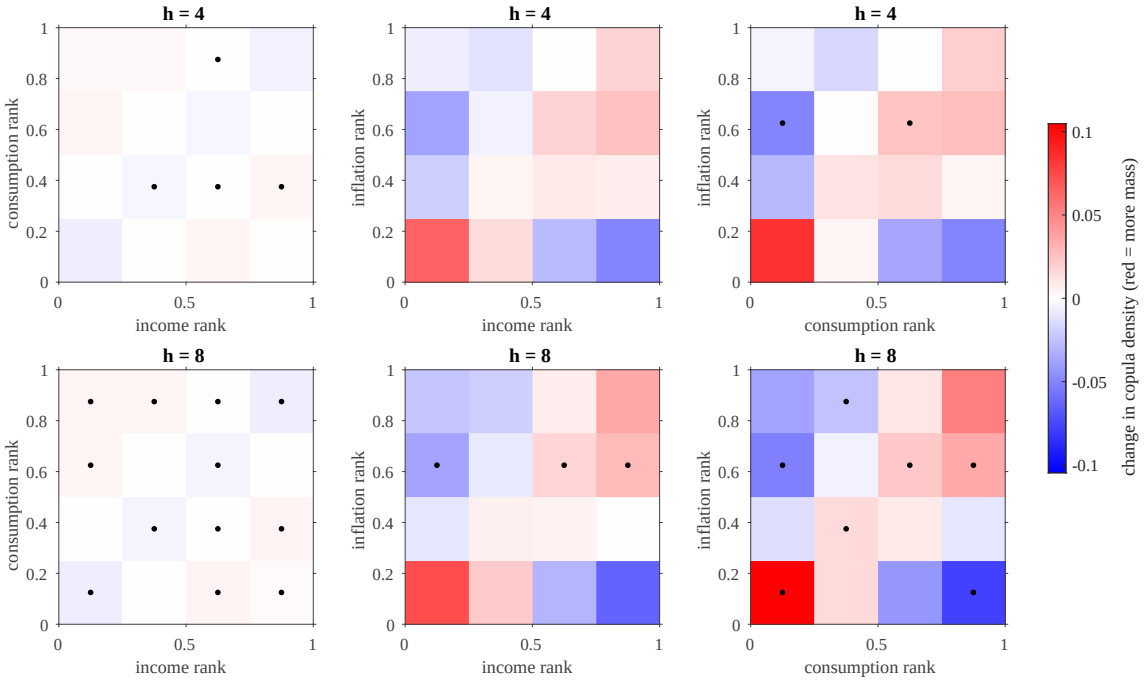


Figure 13: **Where the dependence moves.** Pairwise projections of the response of the copula density (26) at one and two years, obtained by integrating out the third variable. Red means that the shock moves households into that pair of rank cells and blue that it moves them out; dots mark cells whose 68% band excludes zero. All six panels share one colour scale, which is why the income–consumption panel, whose movement is an order of magnitude smaller, appears flat.

D Additional results

D.1 The distribution in levels

Every household variable in the paper is measured relative to its own quarterly mean. The responses in Section 5 therefore describe relative positions, and the effects on levels sit in the macro block. The two can be put back together by re-estimating the three models with one extra variable. The aggregate block of (9) is extended from five variables to six: the growth rate of the distribution's own weighted mean, $100\,\Delta \log \bar{x}_t$, enters next to real GDP growth, aggregate inflation, the gilt yield and the stock market, ordered before uncertainty, which stays last.[14] The shock then moves the level and the shape of the distribution together. The level response of any percentile is its response relative to the mean plus the cumulated response of the mean itself. The extension leaves the relative responses essentially unchanged, and inequality measures are unchanged by construction, since the Gini coefficient and the log spreads do not change when the whole distribution is scaled.

Figure 14 shows what the extension adds: the percentiles of Figure 15 in levels. Two years after the shock, the 5th percentile of the consumption distribution is about 4.4% higher in levels and the 95th about 2.4% lower, with the median about a third of a per cent lower. For gross income, the 5th percentile is about 5.5% higher and the 95th about 0.3% lower. For household inflation, the level picture of the tails is close to the relative one: the 5th percentile falls by about 0.8 and the 95th rises by about 0.9 percentage points within the first year. The compression documented in the main text is therefore not an artefact of normalising by the mean: in levels too, the bottom of both distributions moves up and the top falls back.

The bands are visibly wider than in Figure 15, wide enough to cover zero for most percentiles, and the reason is arithmetic rather than economic. A response in levels is the sum of two estimated pieces: where a household sits within its distribution, and where the distribution itself sits. The second piece is an aggregate whose path becomes harder to pin down the further ahead one looks. The distributional signal is no weaker here; it is simply combined with an aggregate that the data measure less precisely. This is also why the inequality measures of the main text are unaffected: they compare households with each other, so the aggregate cancels out.

D.2 Further figures

The remaining figures support statements made in Section 5, in the order in which they are first mentioned there: the response of individual percentiles, the price series behind the budget-share decomposition, the basket of each housing-tenure group, the Gini responses of the totals and of each component, and the composition of income and of spending along the distribution.

[14] For household inflation the mean is already a rate, so it enters in levels. Everything else is unchanged: the same densities, the same three principal components, the same four lags, the same prior and the same recursive identification, with $n = 9$ variables in place of $n = 8$.

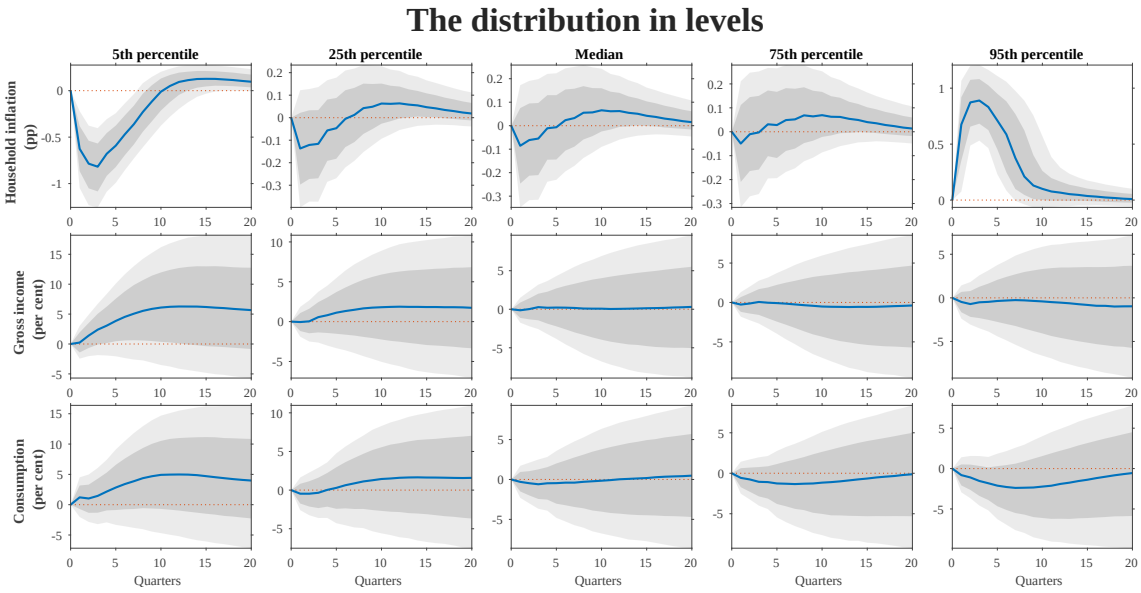


Figure 14: **The distribution in levels.** The levels counterpart of Figure 15: the same rows and the same five percentiles, estimated with the quarterly mean of each distribution added to the VAR as a further aggregate variable, so that the shock moves the level as well as the shape. The level response of a percentile is its response relative to the mean plus the response of the mean itself; inflation is in percentage points, income and consumption in per cent. Posterior medians with 68% and 90% bands. The response of the cross-sectional standard deviation is unchanged by this extension, as are all inequality measures, which are invariant to a common scaling of the distribution.

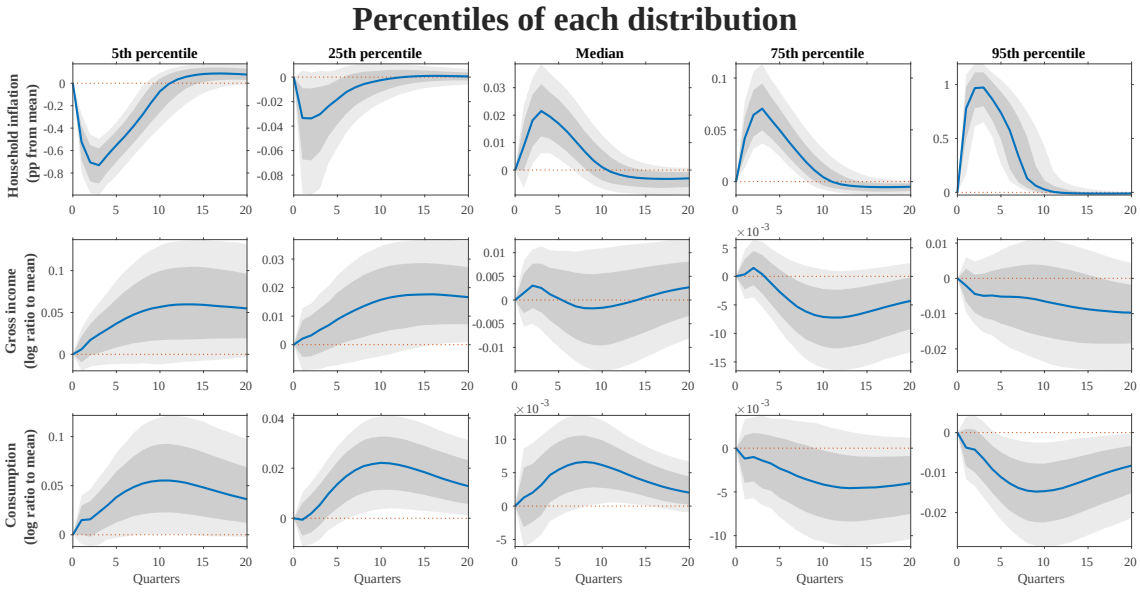


Figure 15: **Percentile responses.** The response of the 5th, 25th, 50th, 75th and 95th percentile of each distribution, by row: household inflation, gross income and consumption. Inflation percentiles are in percentage points, income and consumption percentiles in log points relative to the quarterly mean. Reading across a row shows the same compression or widening that Figure 3 shows as a shift of mass. Posterior medians with 68% and 90% bands.

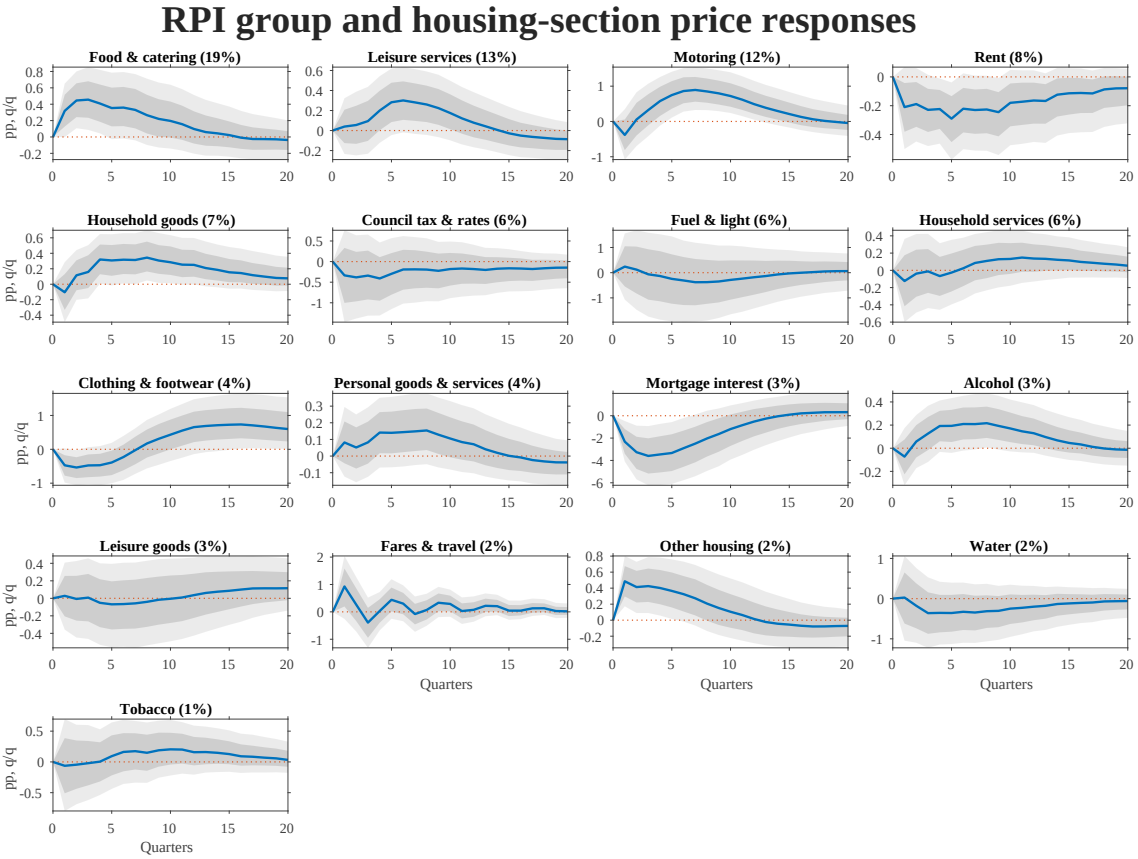


Figure 16: **The seventeen price responses behind the decomposition.** Responses of the twelve RPI commodity group indices and the five housing section indices to the uncertainty shock, ordered by their mean share in household spending (in brackets). These are the series that the budget-share decomposition of Section 4.3 maps through each household's own expenditure shares. Posterior medians with 68% and 90% bands.

E Disposable income

The income distribution of the main text is gross income, before income tax and National Insurance and including benefits. Disposable income, gross income minus direct taxes (Section 3), is estimated in exactly the same way: the same data treatment, macro block, prior and identification, over the same sample, 1977Q1–2018Q1.

Figure 23 shows the density response of disposable income, in the same form as Figures 3 and 7. Its shape is that of gross income in Figure 3: mass leaves the lower tail, around the 5th percentile, and to a much lesser extent the upper tail, and gathers in the lower half and the middle of the distribution, with the picture forming over two to four years. Figure 24 summarises the response by the Gini coefficient and the cross-sectional standard deviation, with the gross-income medians of Figures 18 and 2 dashed. Direct taxes lower the level of inequality, from a baseline Gini of 35.4 for gross income to 32.0 for disposable income, but they barely change its response. The Gini of disposable income falls by close to 0.6 points after four years, against about half a point for gross income, and its 68% band excludes zero from the third year on, as that of gross income does. The

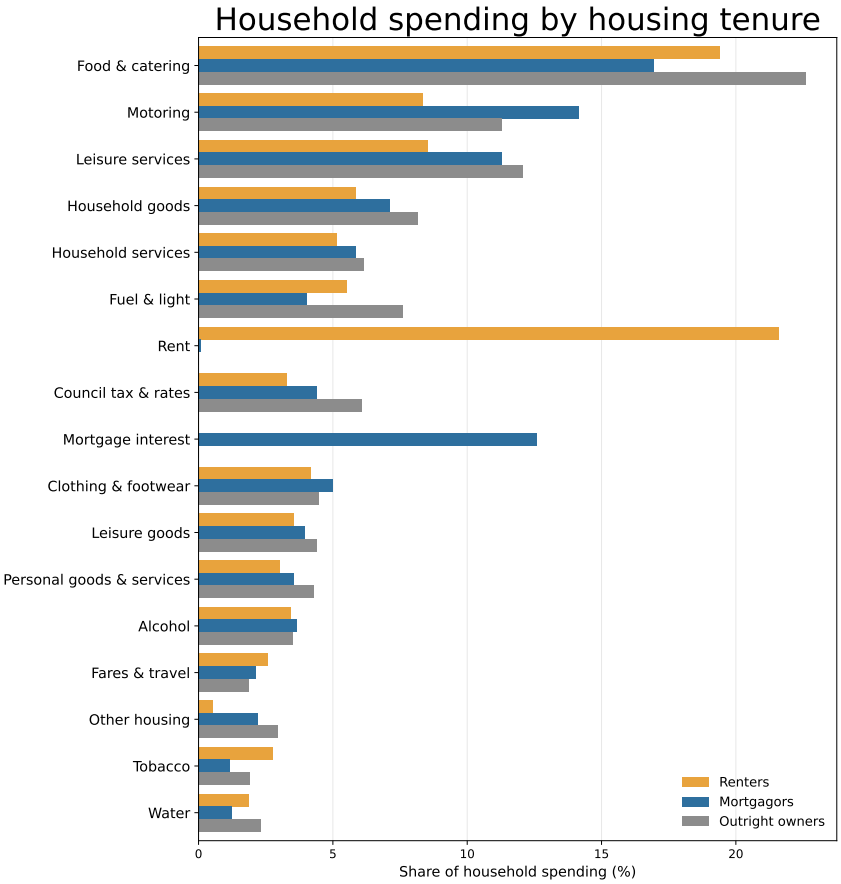


Figure 17: **The basket by housing tenure.** The average share of household spending on each of the seventeen RPI series of the budget-share decomposition, for renters, mortgagors and outright owners, 1992Q1–2018Q1. Shares are survey-weighted means over the households of a group within each quarter, averaged over the quarters, for the households of the tenure comparison in Section 5.3; tenure is defined as in Figure 5. The items are ordered by their average share across all households. Miscellaneous spending, below 1%, is not shown. These are the baskets behind the tenure responses in Figures 4 and 9.

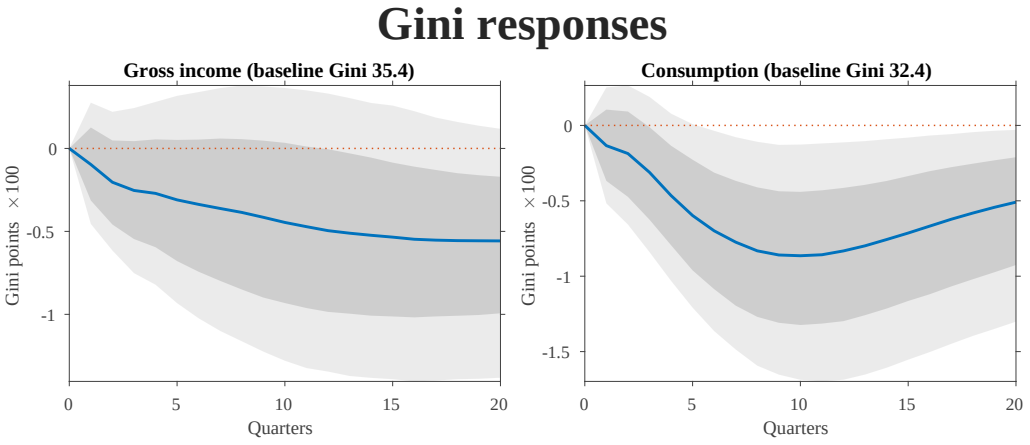


Figure 18: **Gini responses.** The response of the Gini coefficient of gross income and of consumption, in Gini points ×100, computed from the reconstructed densities. The Gini is reported only for these two distributions: household inflation is a percentage-point deviation that takes negative values, for which the coefficient is not defined. Posterior medians with 68% and 90% bands.

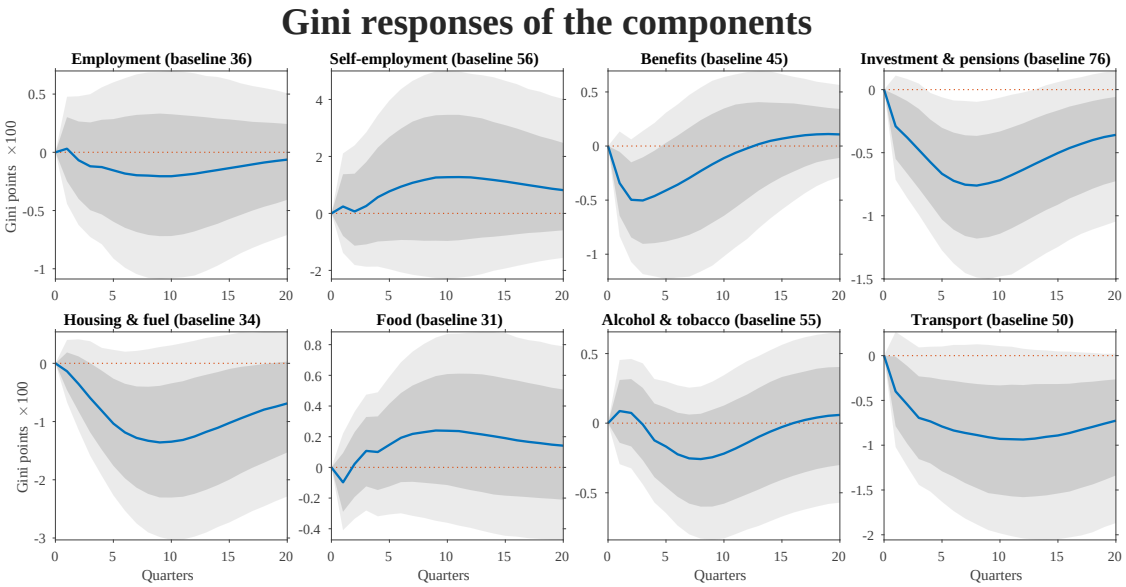


Figure 19: **Inequality within each component.** The response of the Gini coefficient of each income source and each spending category modelled in Figures 7 and 8, with the baseline Gini of that component in brackets. Gini points ×100; posterior medians with 68% and 90% bands.

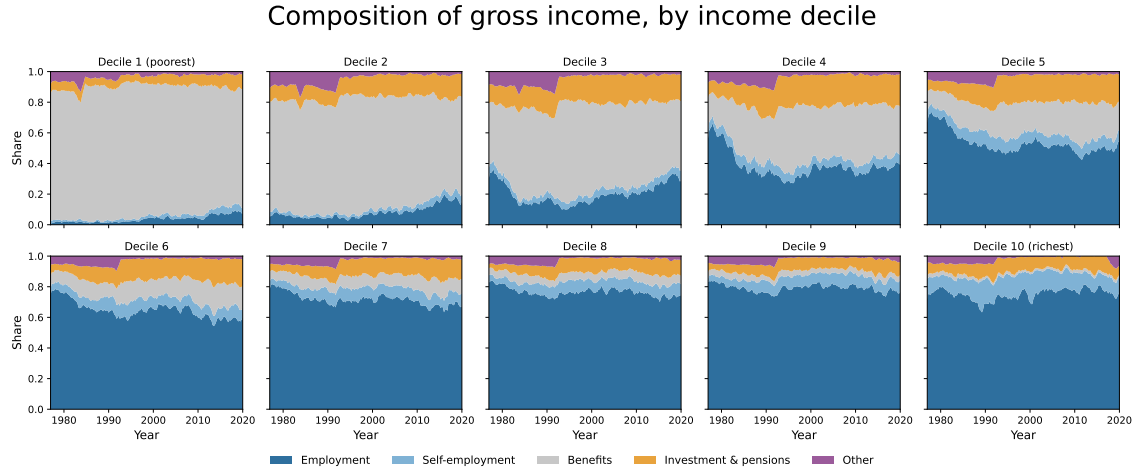


Figure 20: **Where each income source sits in the distribution.** Survey-weighted shares of the five income sources in gross income, within each decile of the gross income distribution, by quarter (four-quarter moving average). Benefits dominate the bottom of the distribution and employment income the middle and the top, while investment and pension income is the largest share of income in the third to fifth deciles and less than a tenth at the top. Deciles are formed on the top-coded income measure used throughout the paper, while the shares are taken over the sum of the five sources, which is the recorded gross income: the Pareto correction of Section 3 adjusts the total of censored observations only and is not allocated to sources, so shares of the corrected total would not add up to one for the top decile in the years since 2008, when about 38% of its households are censored.

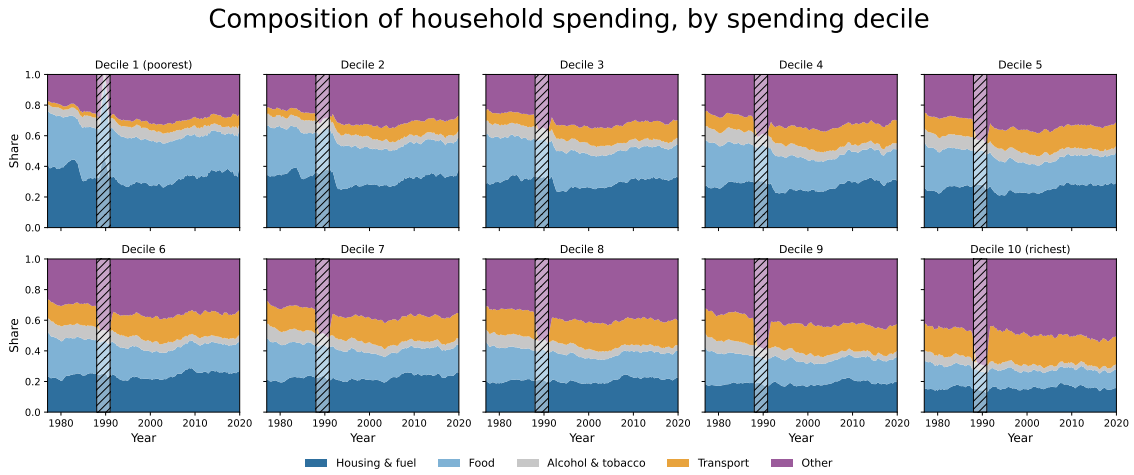


Figure 21: **Where each spending category sits in the distribution.** Survey-weighted shares of the five spending categories in total expenditure, within each decile of the consumption distribution, by quarter (four-quarter moving average). Housing and fuel and food are the two largest items at the bottom and shrink steadily towards the top, where transport and the residual category take their place. The hatched window marks 1988–1990: that vintage of the survey deposits household aggregates for housing, fuel, food, alcohol, tobacco, clothing and a miscellaneous residual only, with no household total for motoring or fares, so transport spending is contained in "Other" for those three years. The only transport variables in those files are person-level transaction records on a different basis, which are not comparable with the household aggregates used here.

standard deviation compresses more slowly at first, by about three quarters of the gross-income response after two years, and is nearly as large after four.

The compression documented in the main text is therefore already present in income before taxes, and the direct-tax system passes it on to disposable income almost unchanged. This is in line with Bilbiie et al. (2025). In Norwegian administrative data, they find that the earnings of high-MPC households are more cyclical than those of other households but their disposable income is not. Part of the reason is capital income, which is very cyclical and accrues mostly to low-MPC savers; part is transfers, and less so taxes, which act as an automatic stabiliser. In the UK data the transfers are already part of gross income: they are the benefits that make up most of the income of the bottom decile (Figure 20) and that sit under the lower tail whose movement drives the compression.[15] Taking out direct taxes therefore adds little to that response.

[15] On another note, this also explains the absent *extensive margin* in the U.K. gross income data considered here. There is no consistent time series of households with zero or negative gross income values in the survey.

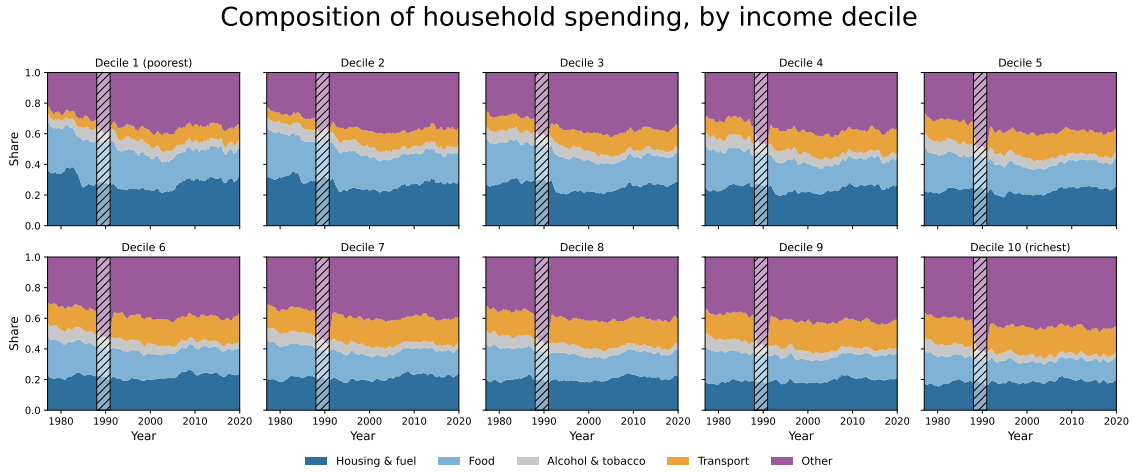


Figure 22: **The basket along the income distribution.** The same spending categories as in Figure 21, and with the same 1988–1990 gap in the transport aggregate, but within each decile of the *income* distribution. This is the gradient that makes an aggregate price index a poor description of any individual household: the poorest decile spends about a quarter of its budget on food and more than a quarter on housing and fuel, while the richest spends a larger share on transport and on the discretionary residual.

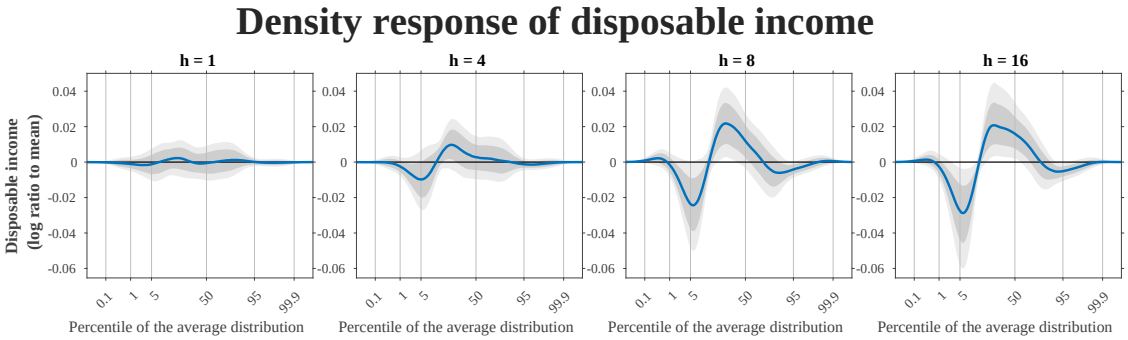


Figure 23: **The response of disposable income.** Functional impulse responses of the distribution of disposable income, gross income minus direct taxes, in the same form as Figure 3: the columns are horizons and the horizontal axis is marked in percentiles of the average distribution. Posterior medians with 68% and 90% bands.

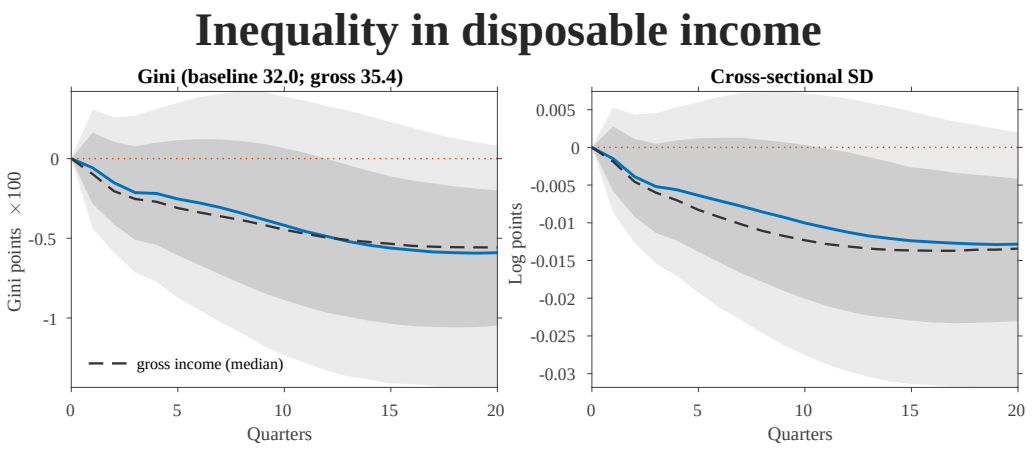


Figure 24: **Inequality in disposable income.** The response of the Gini coefficient (Gini points $\times 100$, left) and of the cross-sectional standard deviation (log points, right) of disposable income, with 68% and 90% bands, and the posterior median of the same statistic for gross income (dashed), as in Figures 18 and 2.